L3 – Modeling to Communicate (UML-lite)

Topics covered

- ✧ Context models
- ✧ Interaction models
- ✧ Structural models
- ✧ Behavioral models

Real Example – Vendor Presentation

✧ How can we better prepare for the presentation?

Subject: Invitation for Interview and Presentation: Potential Vendor Selection

Dear Una team members,

I hope this email finds you well. I am writing to extend an invitation for you to participate in an interview and presentation session regarding the potential collaboration on my project.

After carefully reviewing proposals from various vendors, I believe your company possesses the expertise and capabilities that align closely with the requirements of my project. Your innovative approaches and track record in similar projects make you a strong candidate for this collaboration.

The interview and presentation session will take place on **Tuesday, the 5th March, at 10:00am**. The duration of the interview and presentation will be 30 minutes each. You will have 15 minutes to present your ideas and solutions for our project followed by 15 minutes of questions and discussion. During this session, I invite you to present your ideas for my project and how you plan to execute them effectively. Your presentation should focus on how your proposed ideas will make a significant contribution to the research and align with the project objectives.

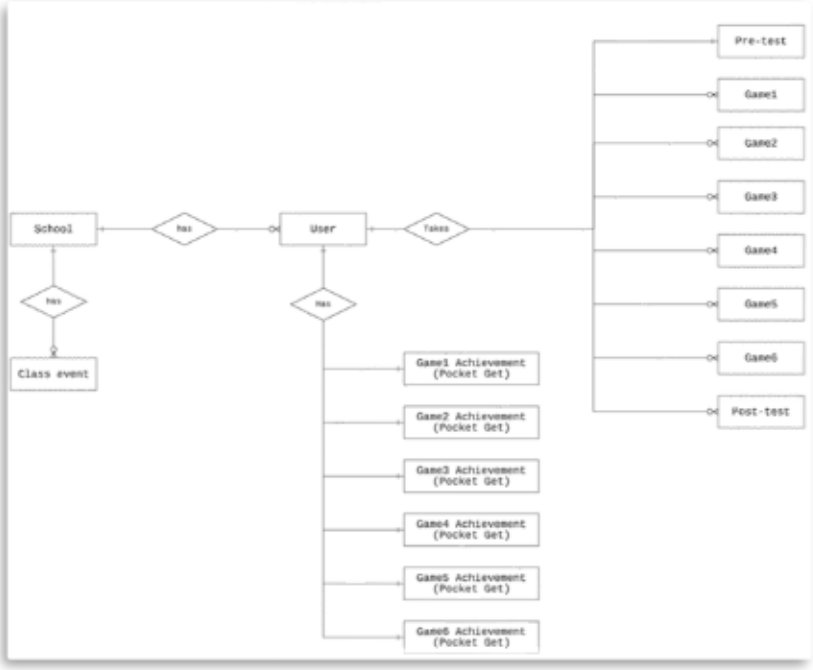
Please come prepared to present and discuss:
Your understanding of the project requirements and objectives.
Specific ideas or strategies you propose to implement.
How your proposed ideas will add value and contribute to the success of the project.
Your team's expertise and resources that will support the execution of the proposed ideas.

The interview will be held online via zoom meeting (TBC), and I kindly request your confirmation of attendance by this **Friday 1st March 2024**. Thank you for your interest in collaborating on this project. I look forward to meeting with you and hearing more about your ideas and capabilities.

Best regards,

System modeling

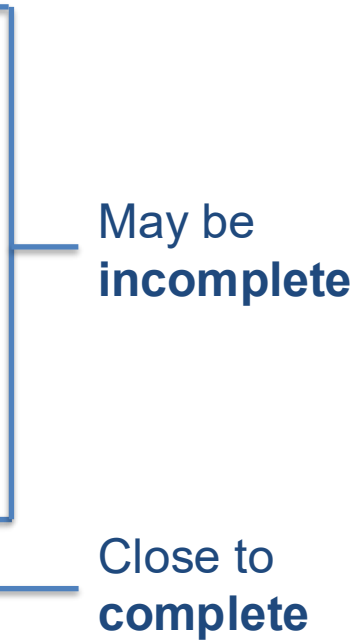
- System modeling is the **process of developing abstract models of a system.**
- Each model presenting a **different view or perspective** of that system.
- Using some kind of **graphical notation**, which is now almost always based on notations in the **Unified Modeling Language (UML)**.
- Helps the analyst to **understand the functionality** of the system and models are used to **communicate** with customers.



Real-life example using Class Diagram, included in the technical report of Making Smart Choice 2, created with LucidChart

Existing and planned system models

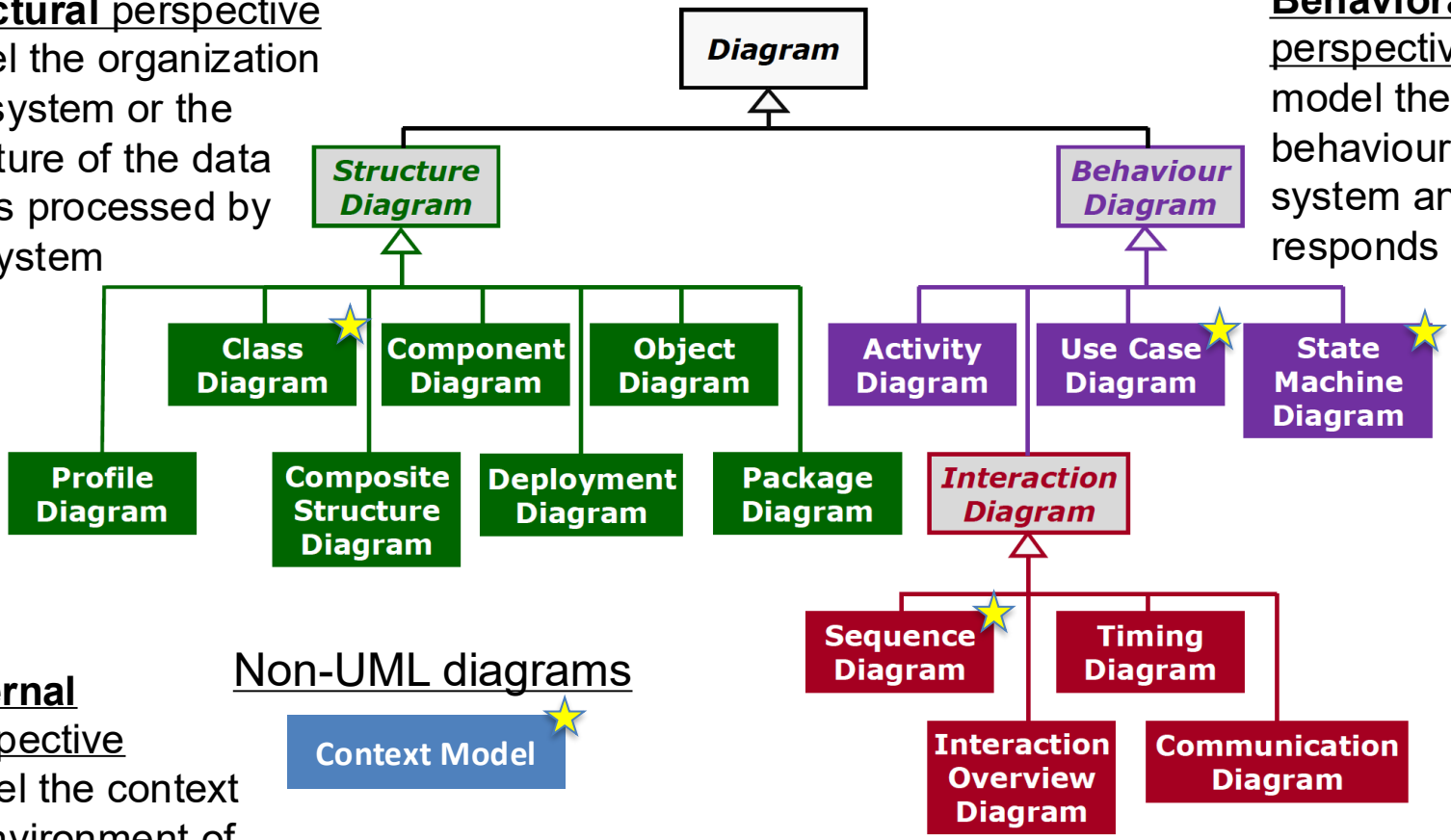
- **Models of the existing system** are used during requirements engineering
  - help clarify what the existing system does and can be used as a basis for **discussing its strengths and weaknesses**. These then lead to **requirements for the new system**.
- **Models of the new system** are used during requirements engineering
  - help **explain the proposed requirements** to other system stakeholders. Engineers use these models to **discuss design proposals** and to **document the system for implementation**.



Classification of UML 2.5 Diagrams

Structural perspective
model the organization of a system or the structure of the data that is processed by the system

Behavioral perspective
model the dynamic behaviour of the system and how it responds to events



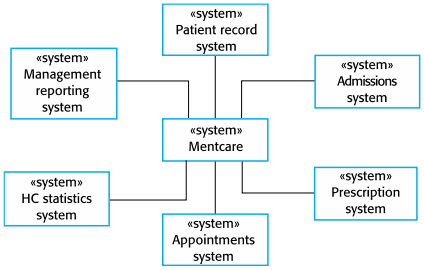
Non-UML diagrams

Context Model

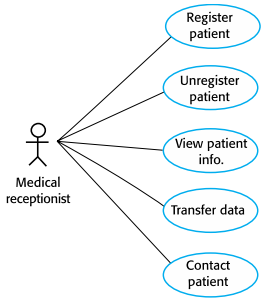
External perspective
model the context or environment of the system

Interaction perspective
model the interactions between a system and its environment, or between the components of a system

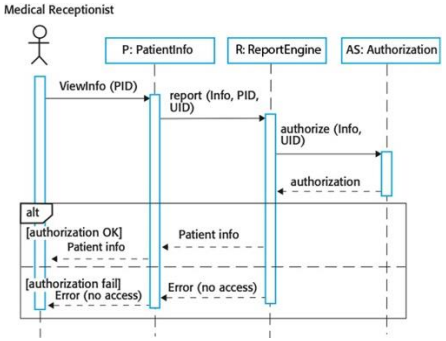
Essential diagram types



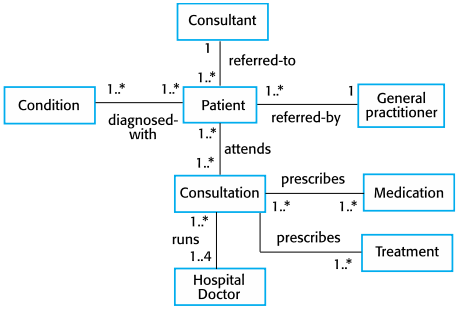
Context diagrams, which show a top-level overview how external entities interact with your system. Ref: (Miro)



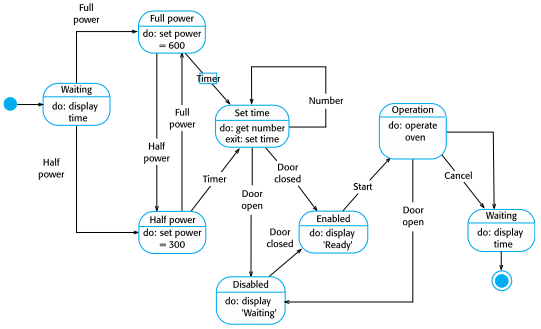
Use case diagrams, which show the interactions between a system and its environment. ***Notation Ref:** (uml-diagrams.org)



Sequence diagrams, which show interactions between actors and the system and between system components. * **Notation Ref**: (uml-diagrams.org)



Class diagrams, which show the object classes in the system and the associations between these classes. * **Notation Ref**: (uml-diagrams.org)

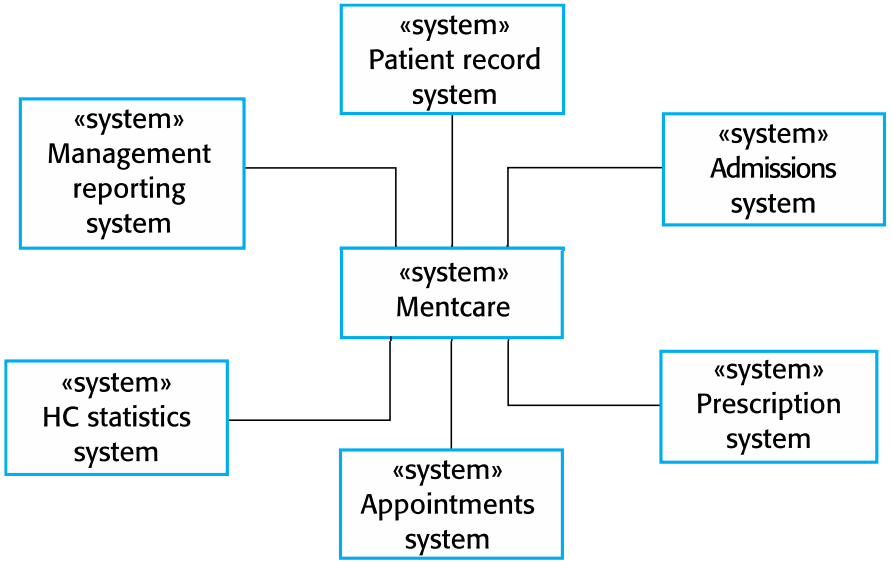


State Machine diagrams, which show how the system reacts to internal and external events. * **Notation Ref**: (uml-diagrams.org)

Context models

Context models

- Illustrate the **operational context of a system** - they show what lies outside the **system boundaries**.
- **System boundaries** are established to define what is inside and what is outside the system.
- **The position of the system boundary has a profound effect** on the system requirements.
- Defining a system boundary is a **political judgment**

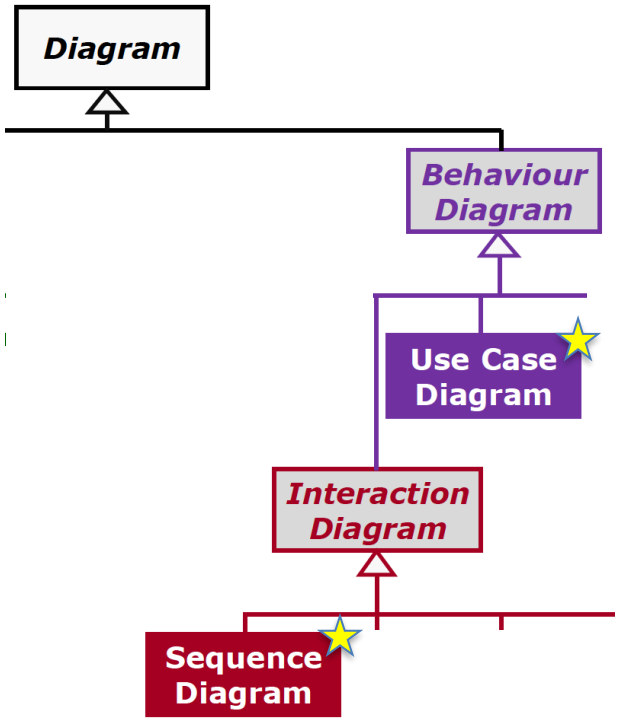


Example: The context of the Mentcare system

Interaction models

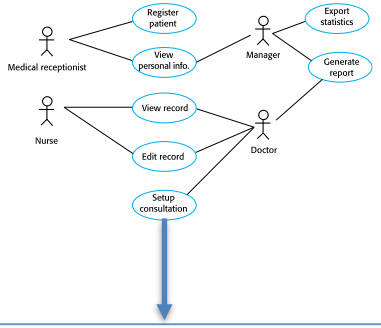
Interaction models

- **Modeling**
  - **user interaction** is important as it helps to identify user requirements.
  - **system-to-system interaction** highlights the communication problems that may arise.
  - **component interaction** helps us understand if a proposed system structure is likely to deliver the required system performance and dependability.
- Use case diagrams and sequence diagrams may be used **together** for interaction modeling.



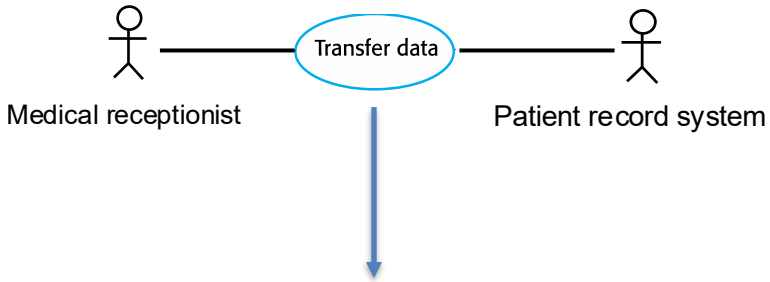
Use case modeling

- Use cases were developed originally to support requirements elicitation and now incorporated into the UML.
- Use case diagrams give **simple overview of interaction**. We need to add more details for complete interaction description. They can be:



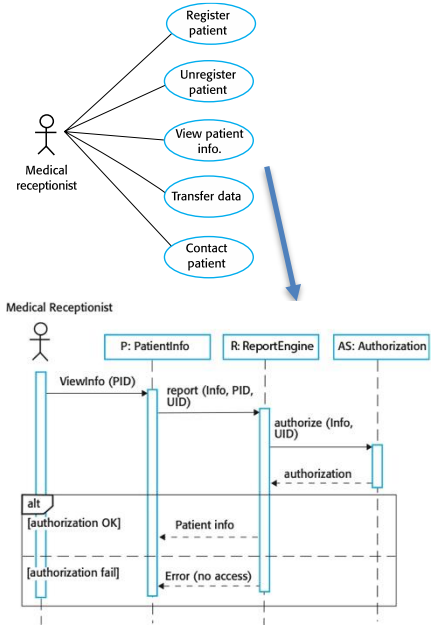
Setup consultation allows two or more doctors, working in different offices, to view the same patient record at the same time. One doctor initiates the consultation by choosing the people involved from a dropdown menu of doctors who are online. The patient record is then displayed on their screens, but only the initiating doctor can edit the record. In addition, a text chat window is created to help coordinate actions. It is assumed that a phone call for voice communication can be separately arranged.

a simple textual description;



Mentcare system: Transfer data	
Actors	Medical receptionist, patient records system (PRS)
Description	A receptionist may transfer data from the Mentcare system to a general patient record database that is maintained by a health authority. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment.
Stimulus	User command issued by medical receptionist
Response	Confirmation that PRS has been updated
Comments	The receptionist must have appropriate security permissions to access the patient information and the PRS.

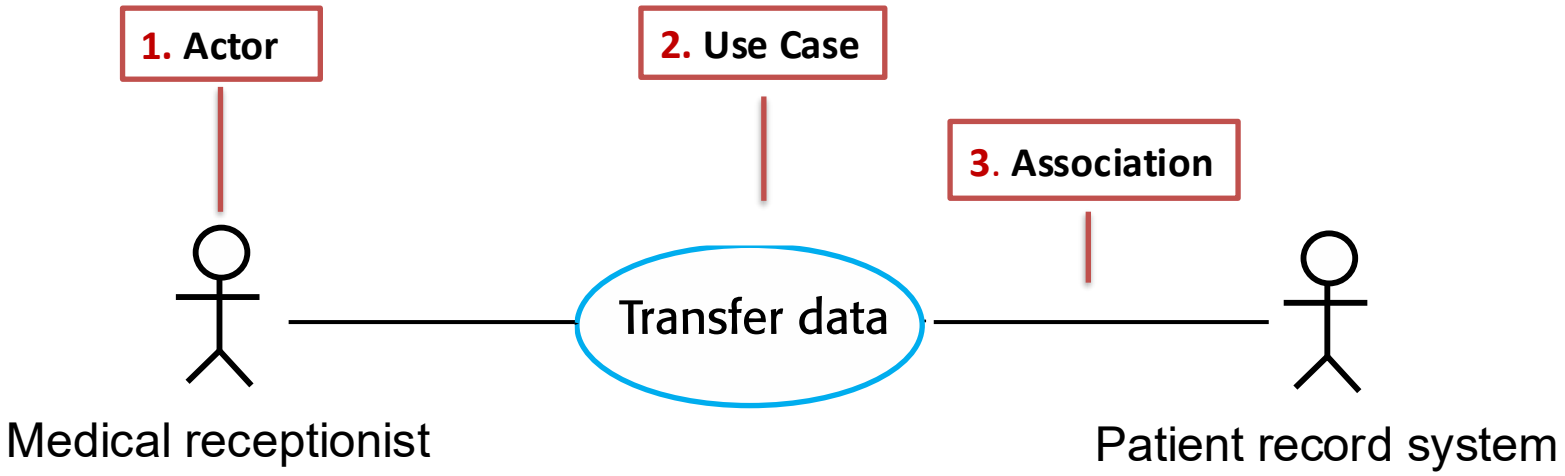
a structured description in a table; or



a sequence diagram

What do the UML notations mean?

- **Actors** in a use case may be **people or external systems**.
- Each **use case** represents a **discrete task** that involves external interaction with a system.

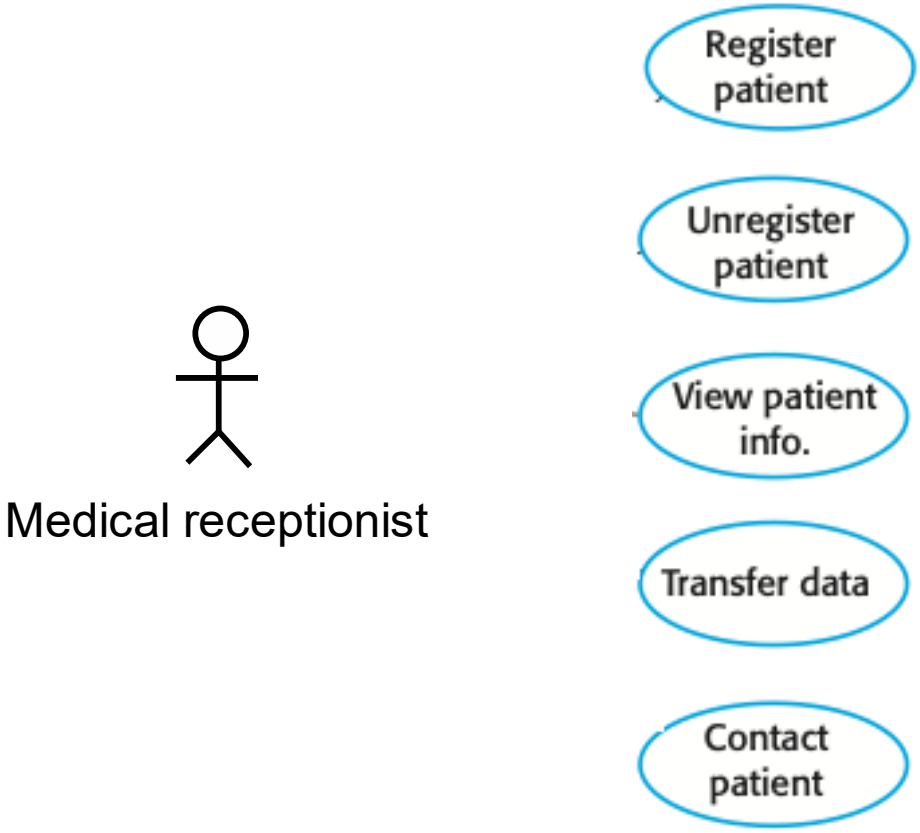


Example: The 'Transfer Data' use case in the Mentcare system

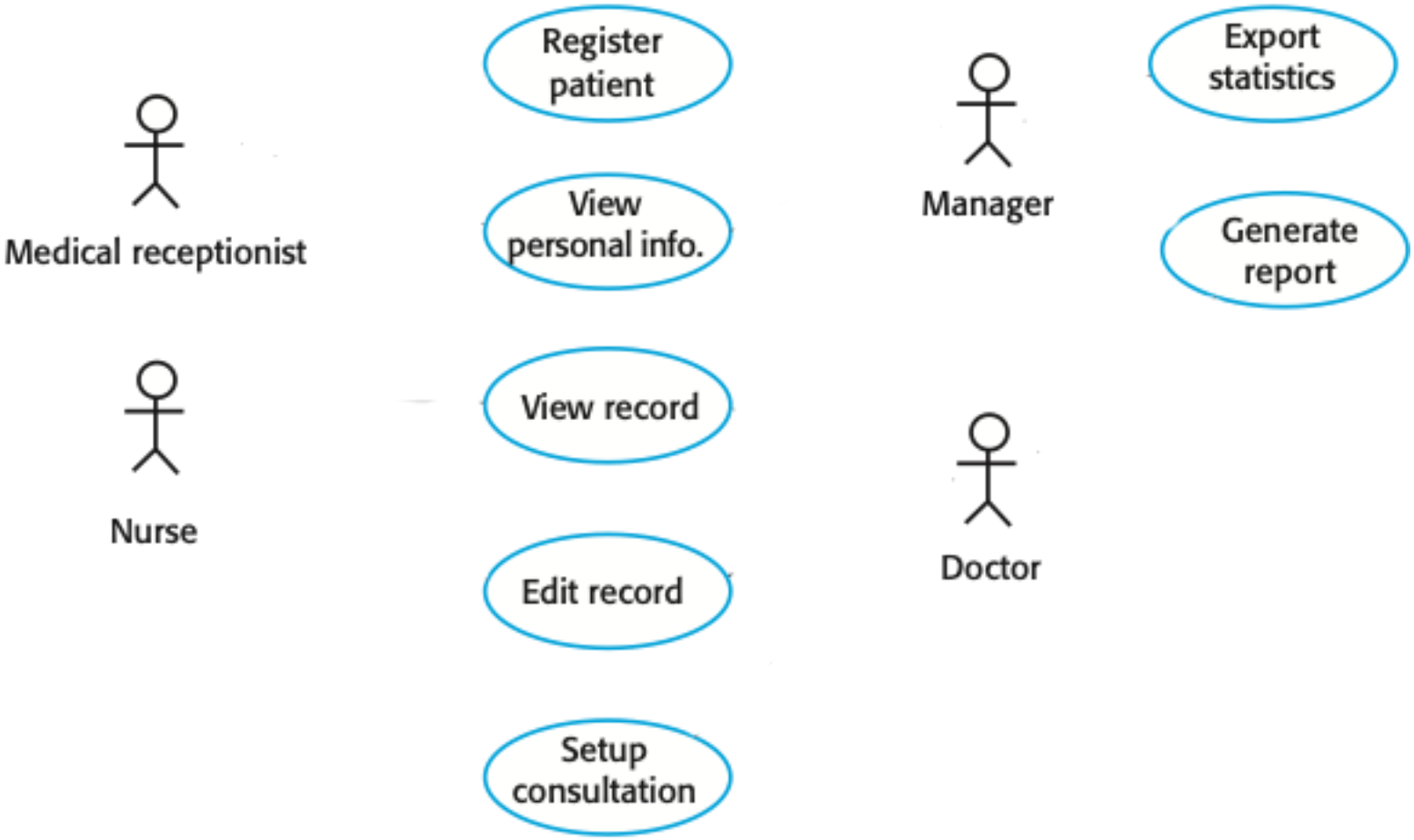
Common question 1: Is it possible to use an arrow instead of a line?



Common question 2: Is it possible to have multiple use cases within a system?



Common question 3: Can a system have multiple use cases and multiple actors?



The ‘Transfer data’ model should align with the description of the ‘Transfer data’ use case

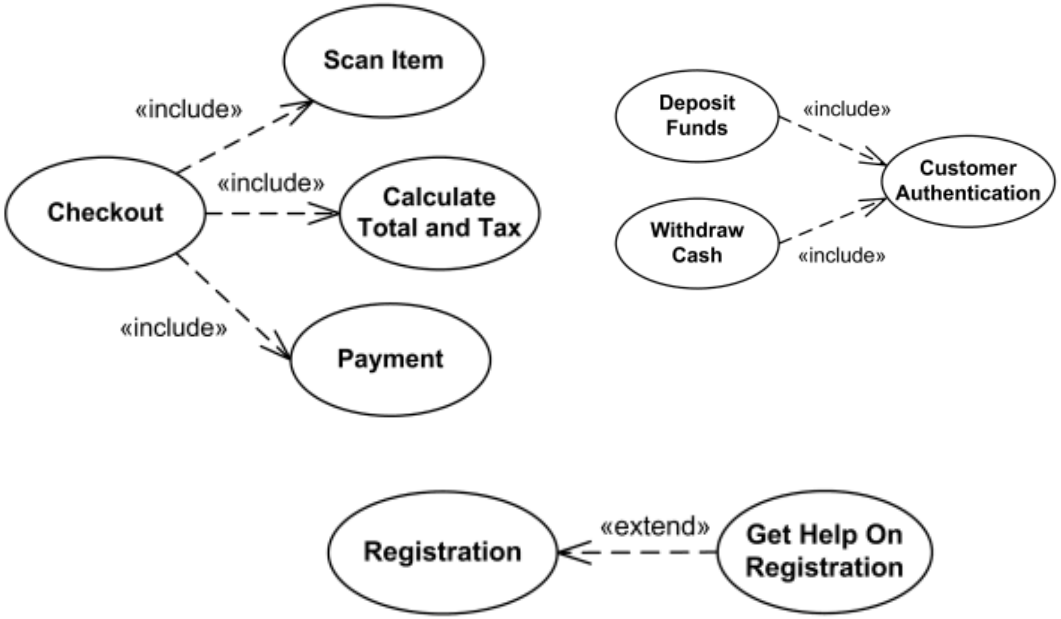


Mentcare system: Transfer data	
Actors	Medical receptionist, patient records system (PRS)
Description	A receptionist may transfer data from …….. or a summary of the patient’s diagnosis and treatment.
Stimulus	User command issued by medical receptionist
Response	Confirmation that PRS has been updated
Comments	The receptionist must have appropriate security permissions to access the patient information and the PRS.

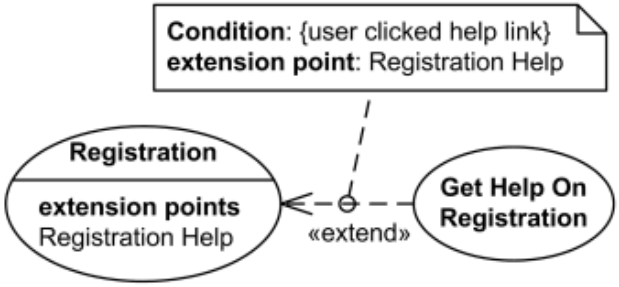
Example: Tabular description of the ‘Transfer data’ use case

Recommended Readings

- **Include** (uml-diagrams.org) and **extend** (uml-diagrams.org)

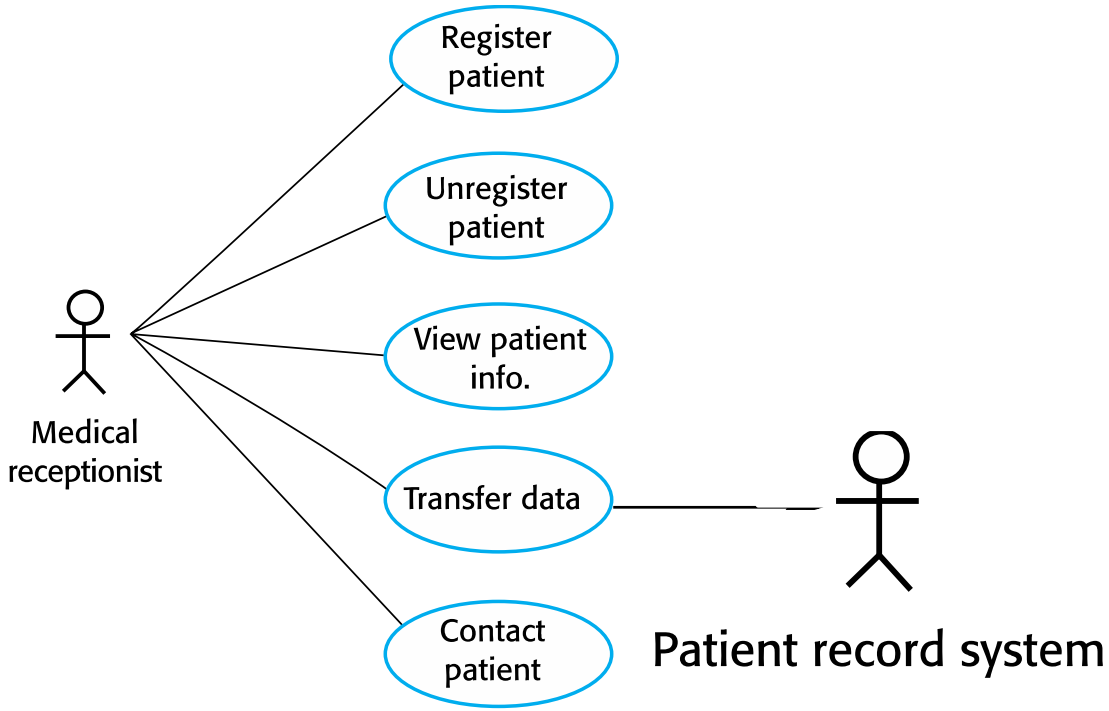


- Way to add **comments**



The use case diagram for the Mentcare system, including all use cases

- Overview of interaction of **each use case** can be presented by **a sequence diagram**



(See two examples in the next few slides)

Sequence diagrams

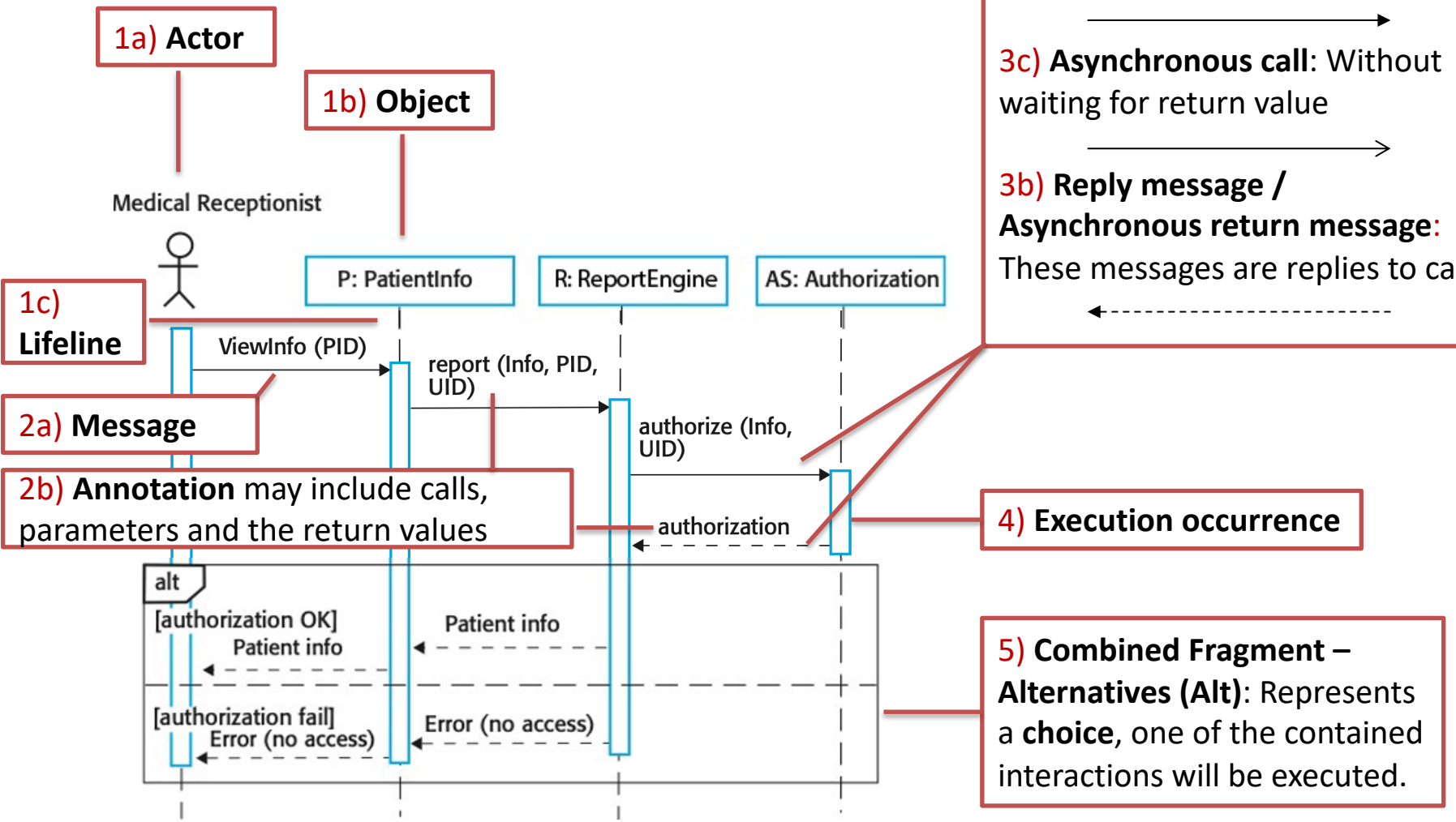
- UML sequence diagrams model the **interactions between the actors and the objects** within a system.
- A sequence diagram shows the sequence of interactions that take place during **a particular use case** or **use case instance**.

Mentcare system: View Patient Information	
Actors	Medical receptionist
Description	A medical receptionist requests to view patient information. The request is processed through the report engine, which verifies the receptionist's authorization to access the information via the authorization component. If authorization is granted, the patient information is displayed; otherwise, an error message is returned.
Stimulus	ViewInfo command issued by the medical receptionist
Response	Display of patient information or an error message indicating no access
Comments	The receptionist must have appropriate security permissions to access the patient information. Authorization is required to ensure secure access.



Sequence Diagram

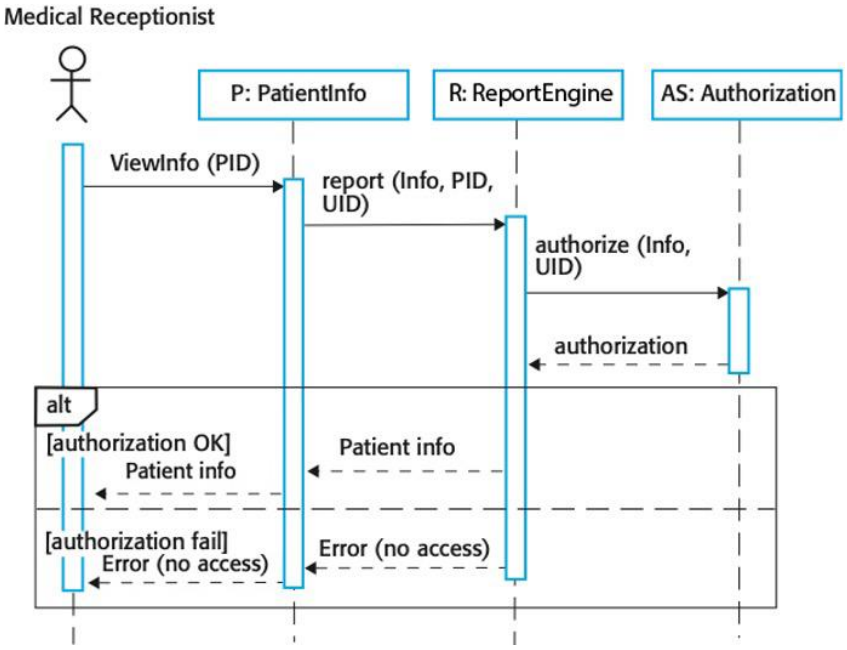
Develop the sequence diagram from the description, step by step.



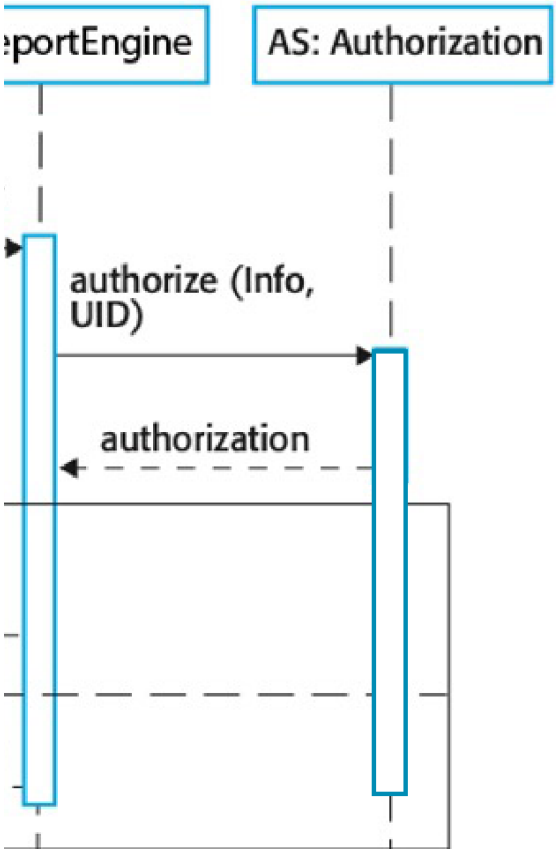
Example: Sequence diagram for 'View Patient Information'

Does the sequence diagram align with the description?

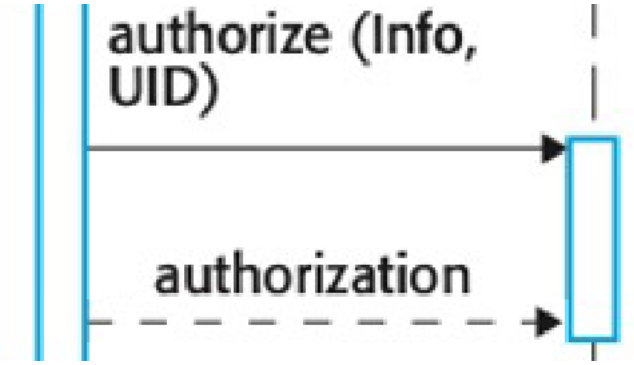
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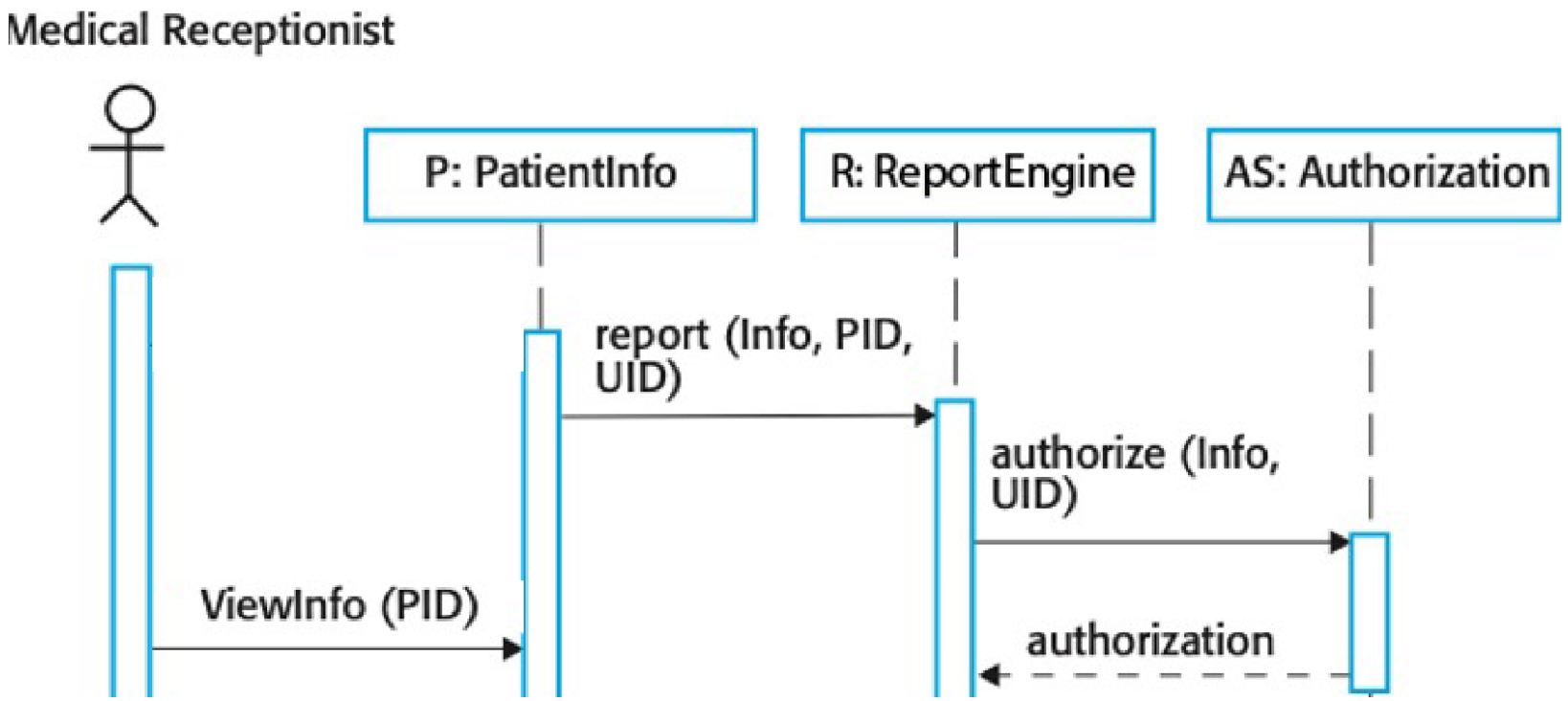
Any problem(s)?



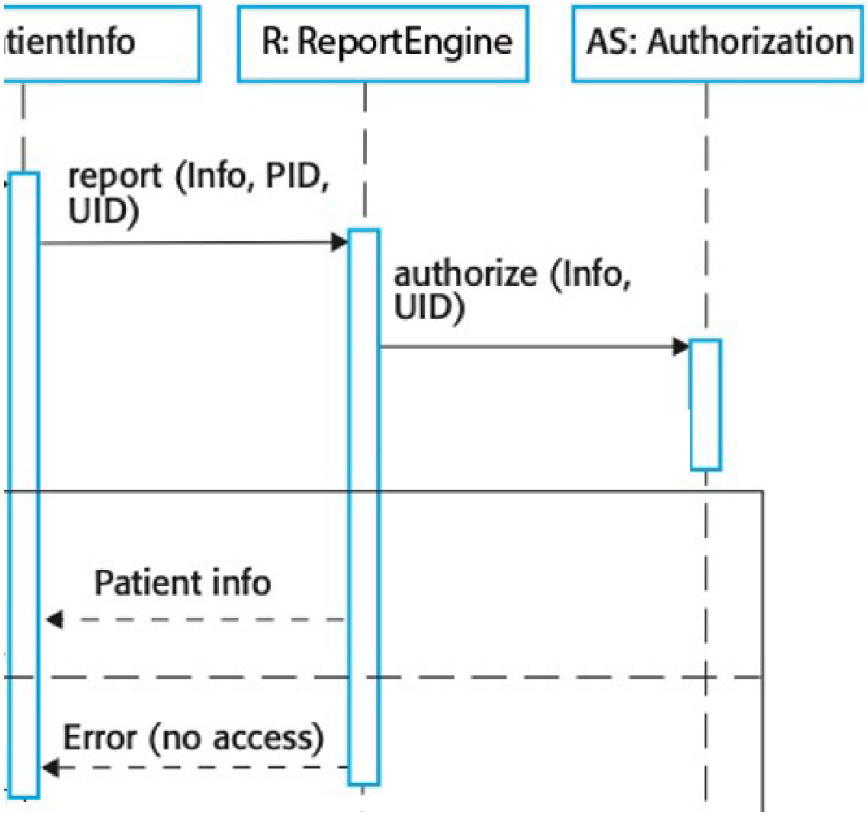
Any problem(s)?



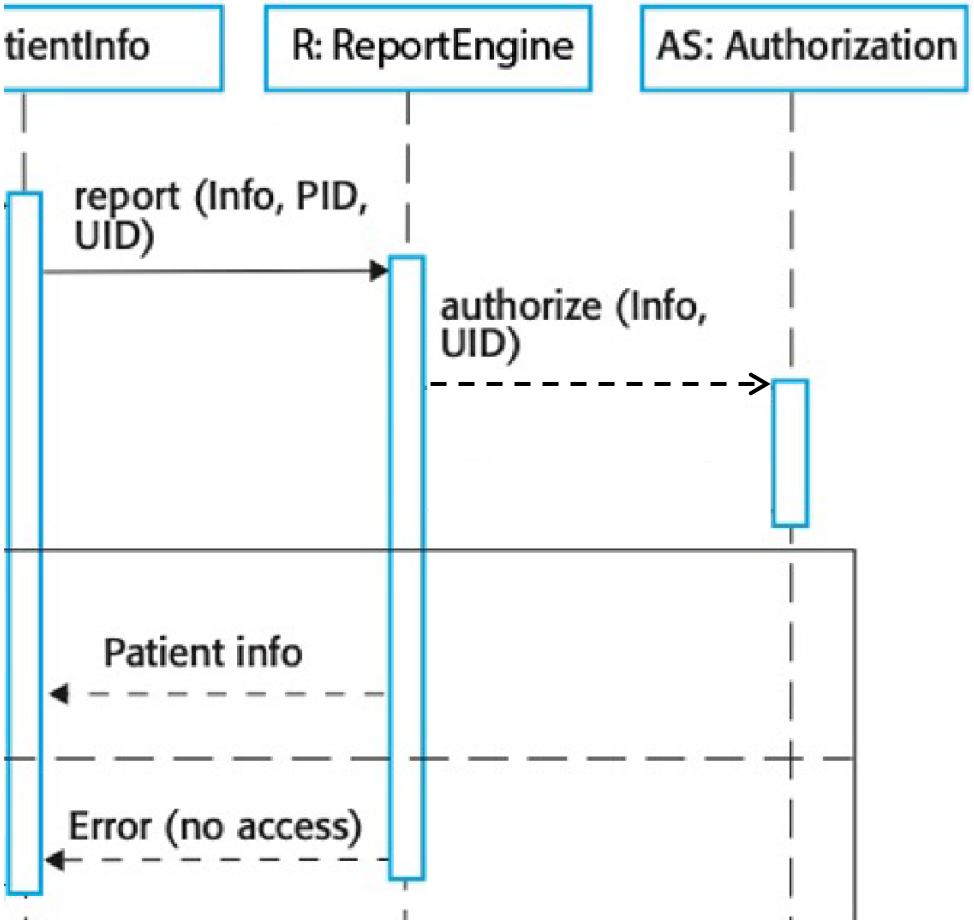
Any problem(s)?



Any problem(s)?



Any problem(s)?



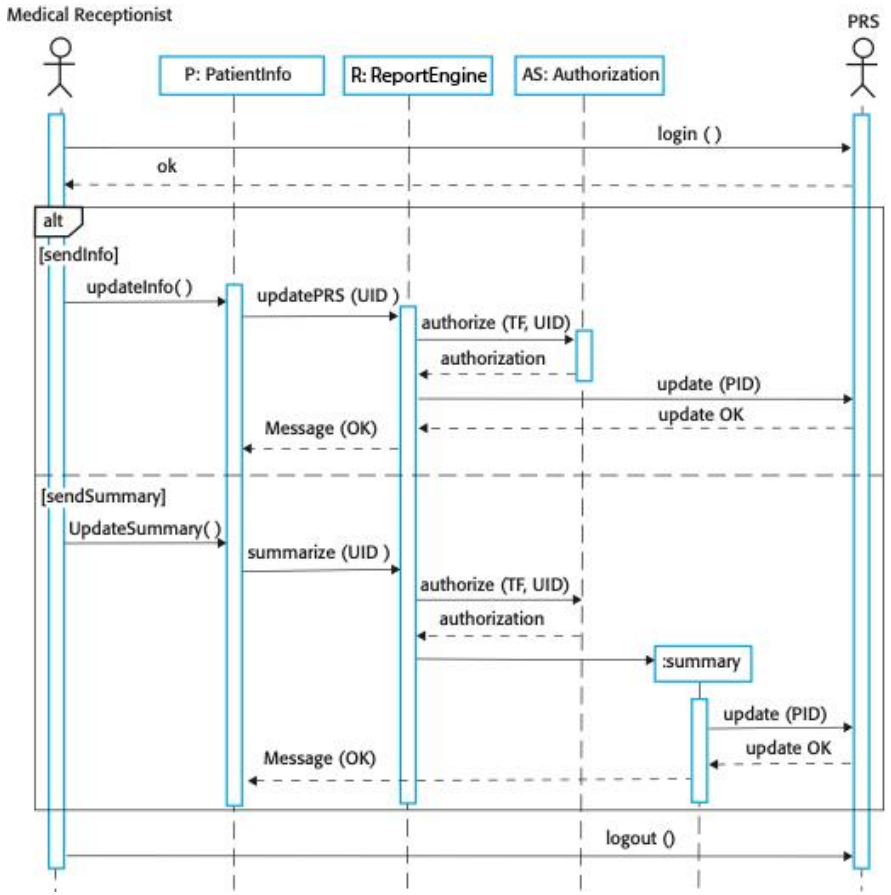
Presentation

To test your knowledge, try to present the interactions of this sequence diagram.

Hint:

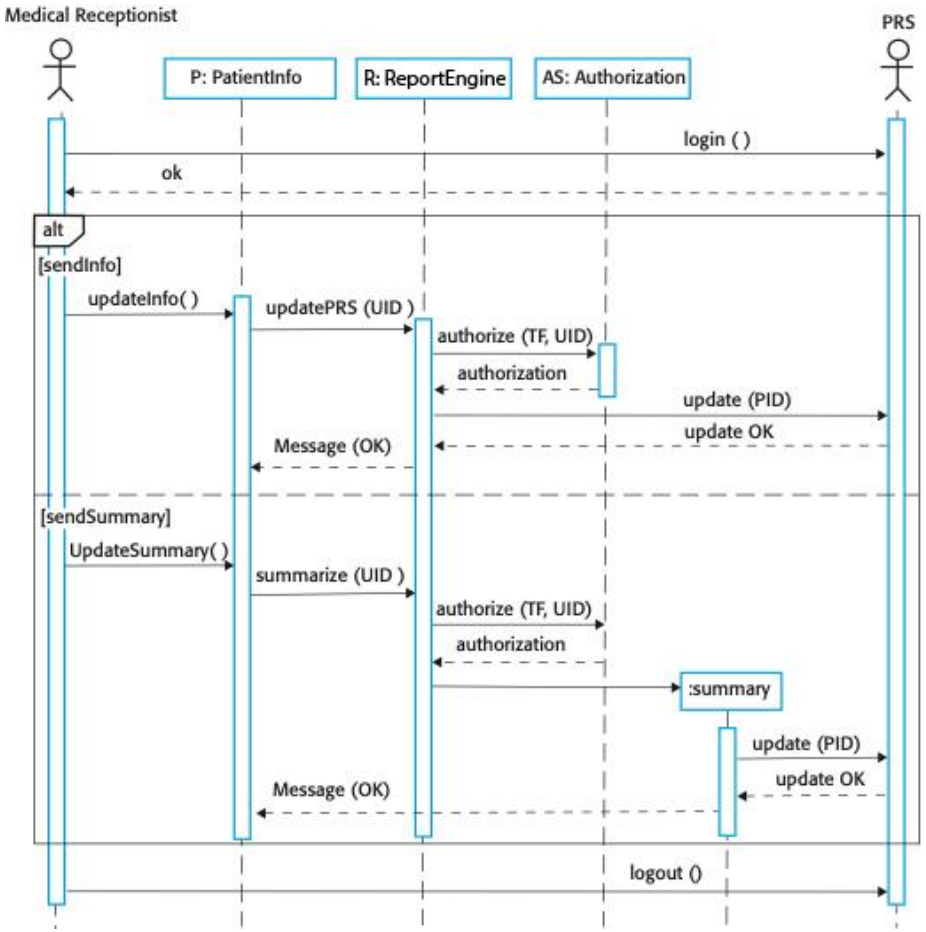
1) Any common interaction(s)?

2) Any extra object(s)?



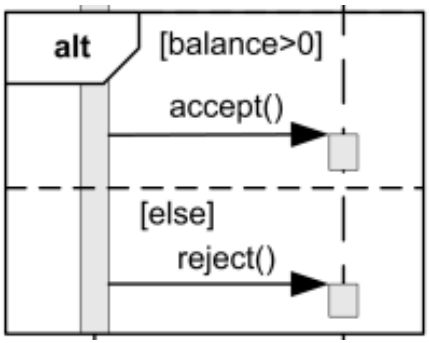
Example: Sequence diagram for 'Transfer Data'

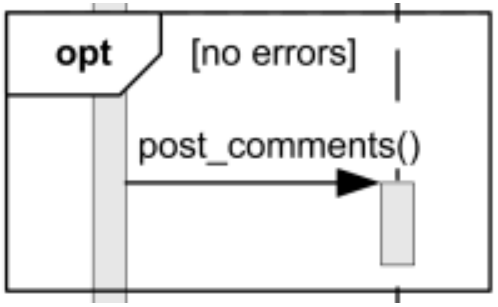
Any UML notation-related improvement(s) from the Sequence Diagram in the previous slide?

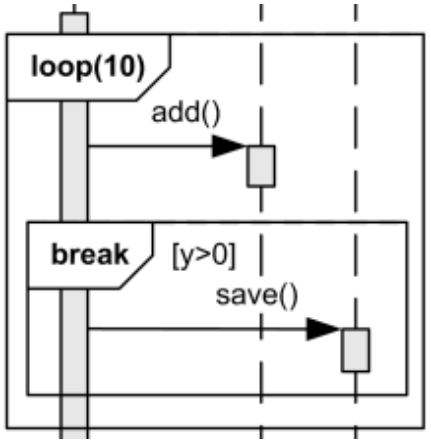


Required Readings

- **Combined fragments**, e.g. alt, opt, loop, break (uml-diagrams.org)



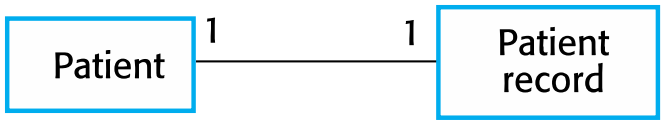




Structural models

Structural models and Class diagrams

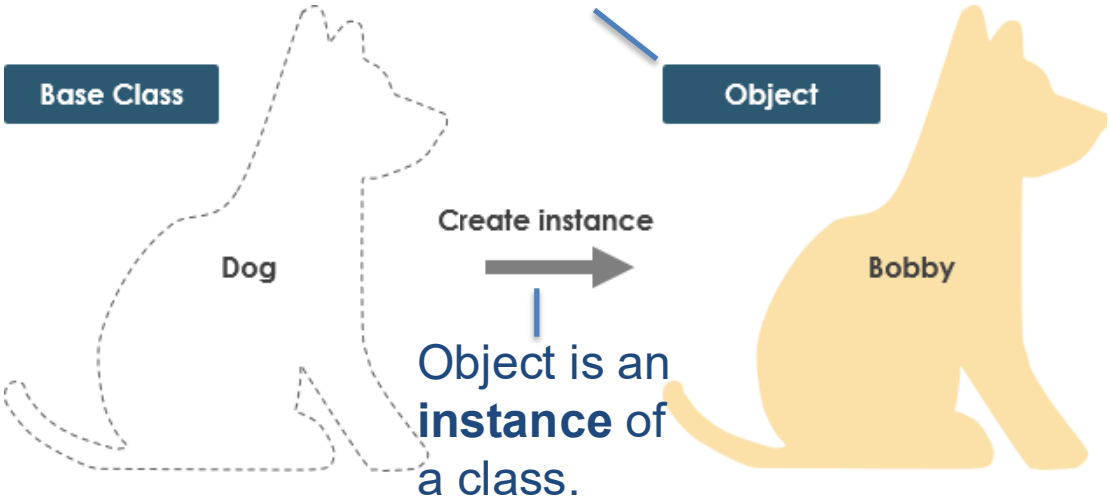
- Structural models
  - **display the organization of a system** in terms of the components that make up that system and their **relationships**.
  - may be **static** models, which show the structure of the system design, or
  - may be **dynamic** models, which show the organization of the system when it is executing.
- Class diagrams
  - are used when developing an **object-oriented system** model to show the classes in a system and the associations between these classes.



What is a Class?

A **Class** is a blueprint for an object.

Each **Object** was built from the same set of blueprints and therefore contains the same **properties** and **methods**.



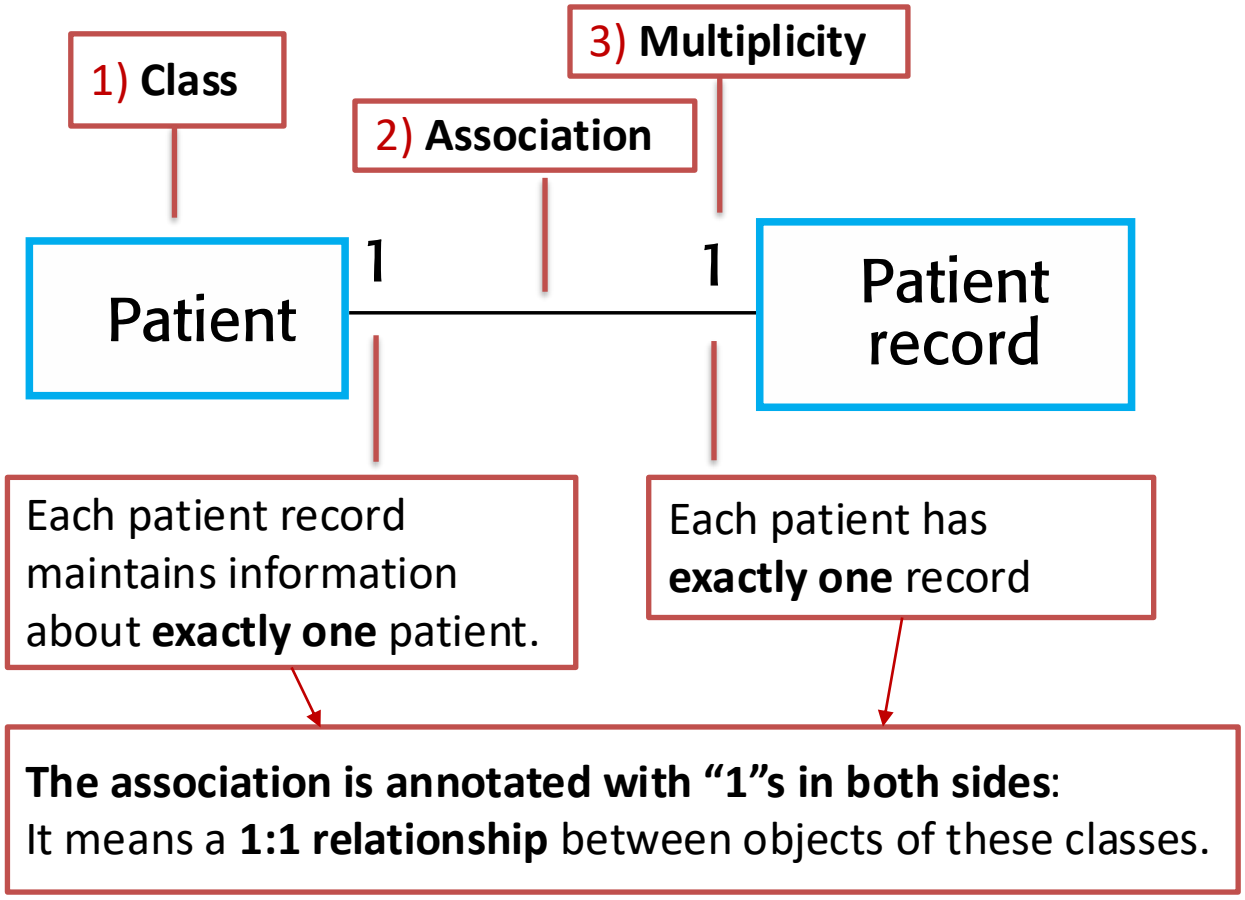
(Ref: Visual Paradigm)

Class and Instance in Sequence Diagram

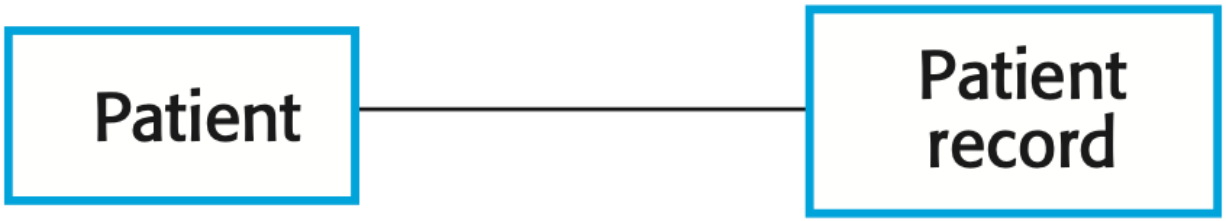


- 'P' is the **instance** of the **class** 'PatientInfo'
- 'R' is the **instance** of the **class** 'ReportEngine'
- 'AS' is the **instance** of the **class** 'Authorization'

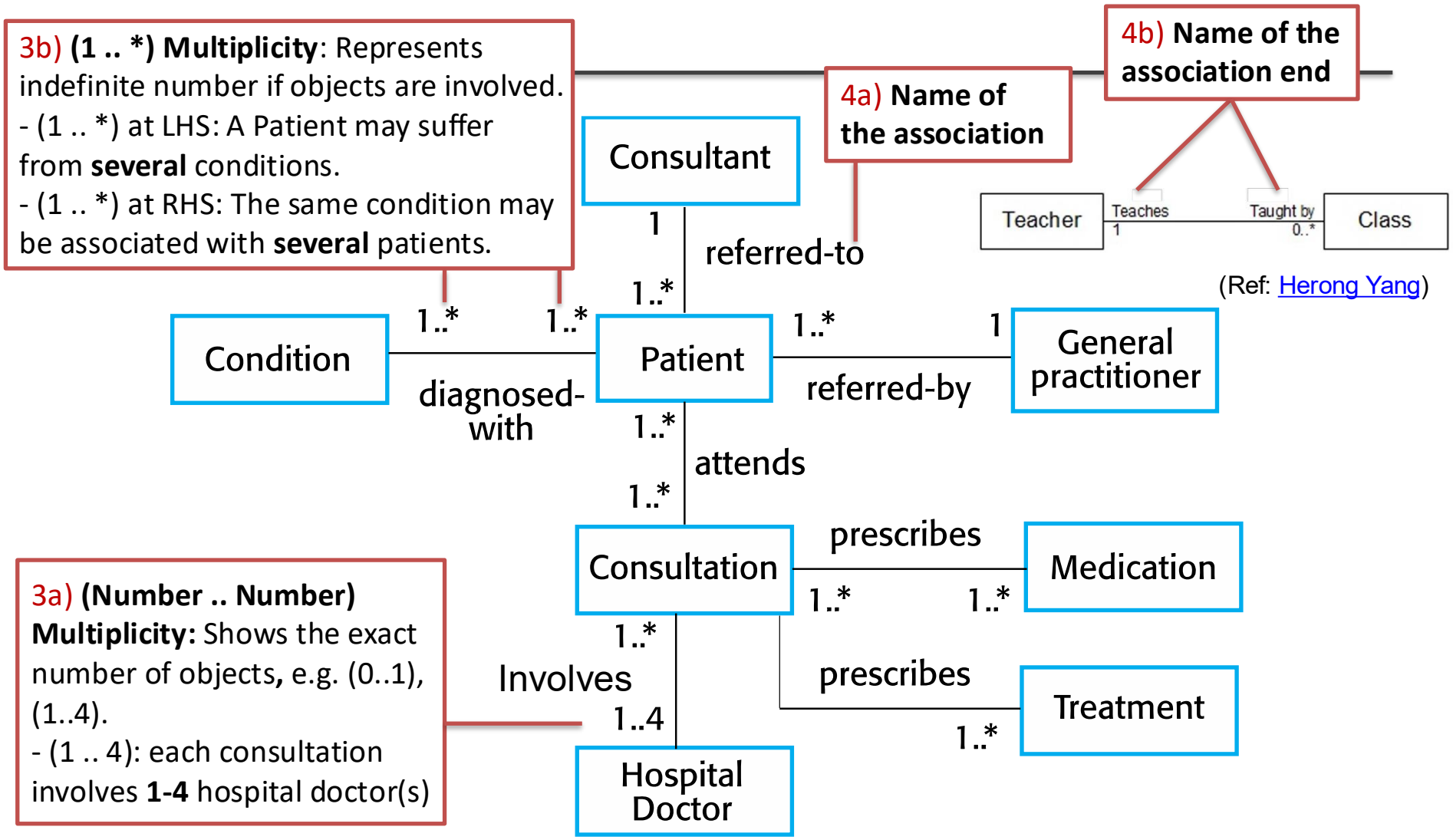
What do the UML notations mean?



Any problem(s)?

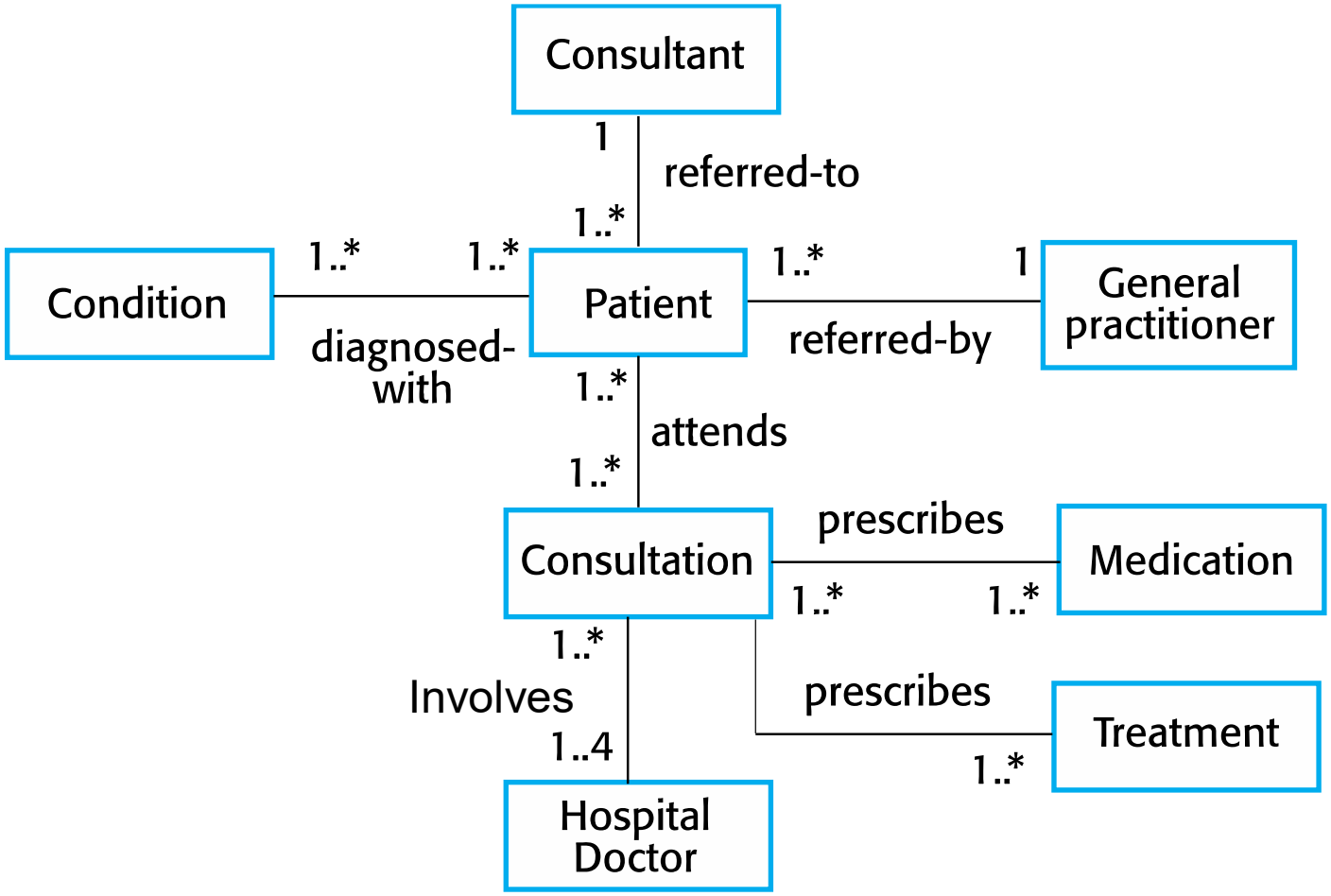


What do the UML notations mean?



Example: Classes and associations in the MHC-PMS

Any problem(s)?

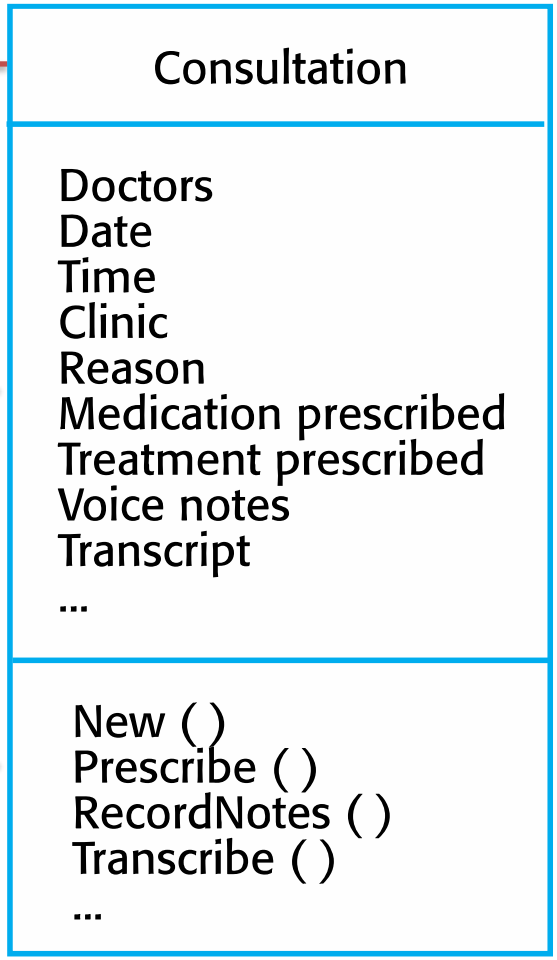


What do the UML notations mean?

1) **Class Name**

2) **Class Attributes**

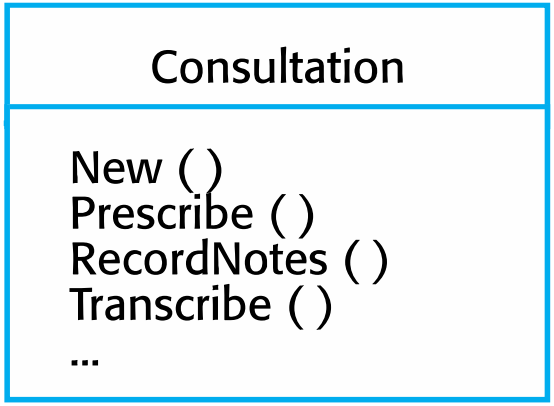
3) **Class Operations/Methods**





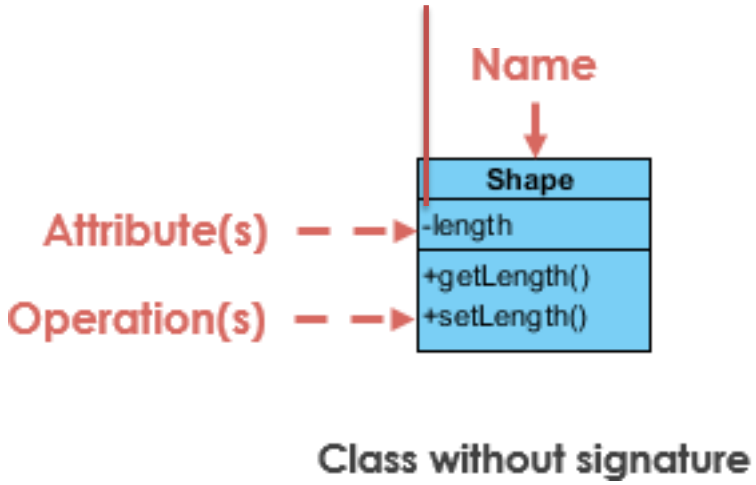
Example: The Consultation class

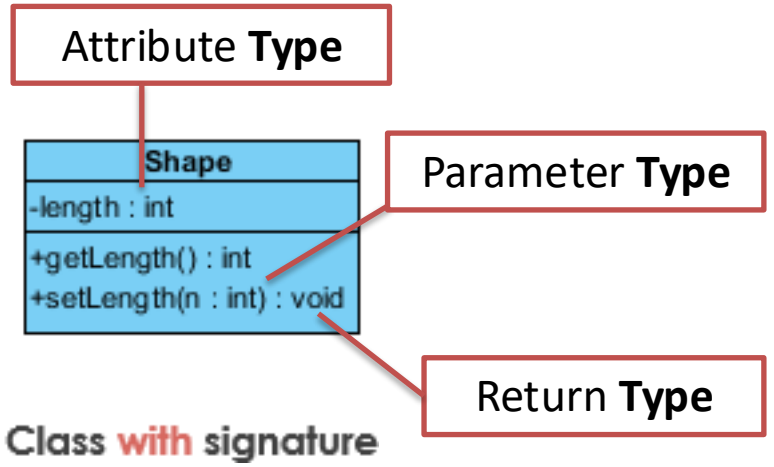
Any problem(s)?



UML Class Notation and Visibility

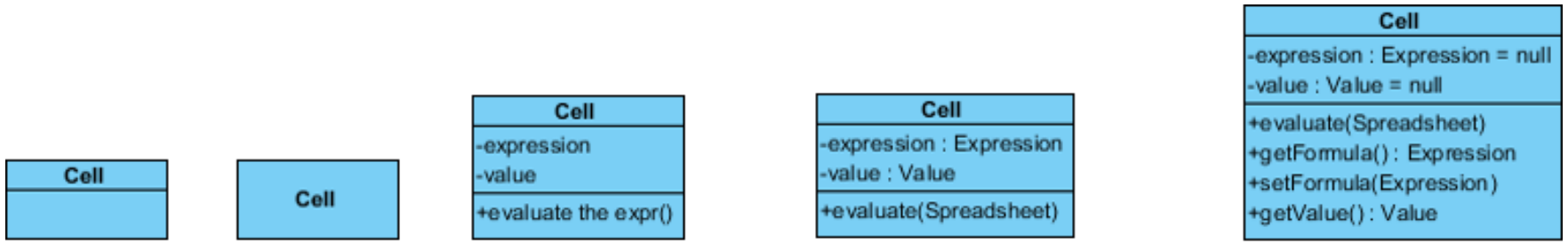
Visibility
+ denotes **public** attributes or operations
- denotes **private** attributes or operations





Perspectives of Class Diagram

- Class diagram can be interpreted from various perspectives:



Conceptual

represents the concepts in the domain

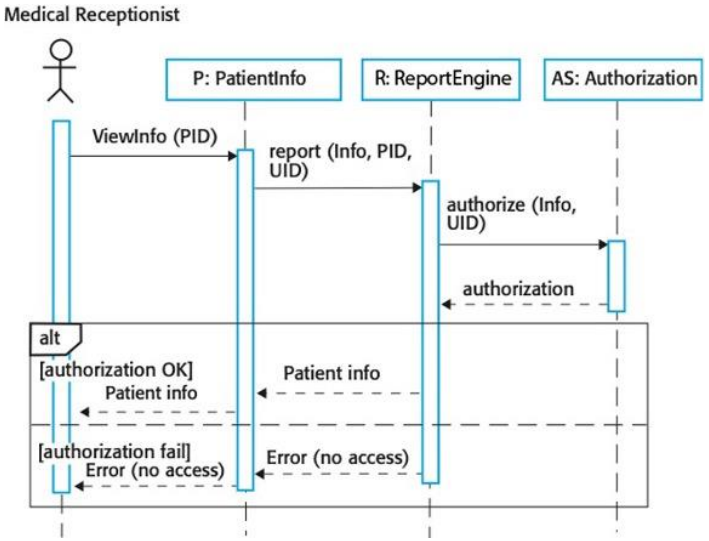
Specification

focus is on the interfaces of Abstract Data Type (ADTs) in the software

Implementation

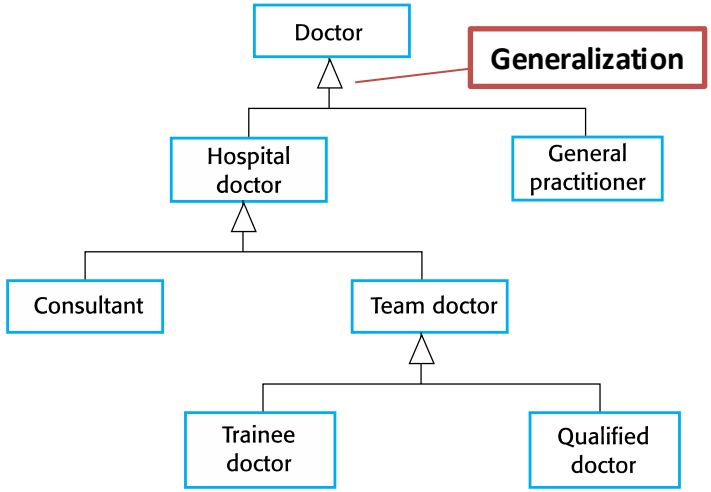
describes how classes will implement their interfaces

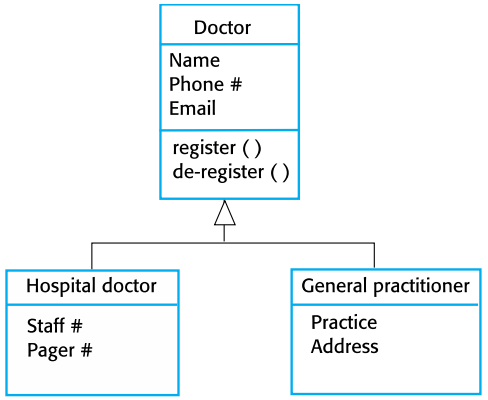
Develop the class diagram from the sequence diagram, step by step. Let's try!



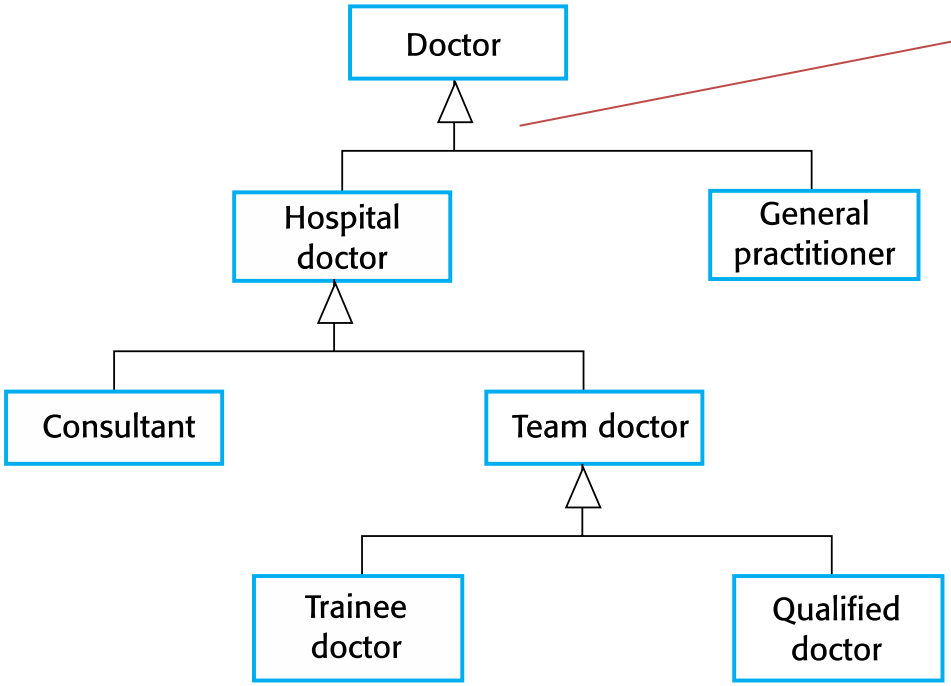
Generalization

- Generalization **manages complexity**
- Infers that **different members** of these classes (e.g. Hospital doctor, General practitioner) have some **common characteristics** (e.g. Doctor)
- **Advantage**: If changes are proposed, then you do not have to look at all classes in the system to see if they are affected by the change.
- In object-oriented languages, such as Java, generalization is **implemented** using the **class inheritance mechanisms** built into the language.





What do the UML notations mean?



Generalization – pointing up to a more general class. It forms an **"is-a"** relationship.

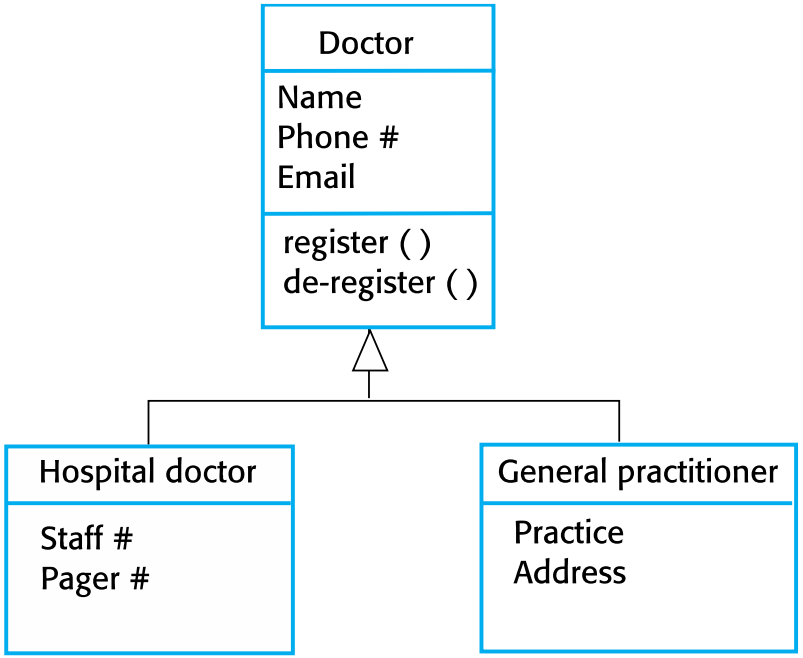
In this example,

a) General practitioners and hospital doctors can be generalized as doctors

b) There are three types of Hospital Doctor:

   I. Trainee Doctor
   II. Qualified doctor
   III. Consultant

What do the UML notations mean?



The lower-level classes are **subclasses** that **inherit** the attributes and operations from their **superclasses**.

In this example,

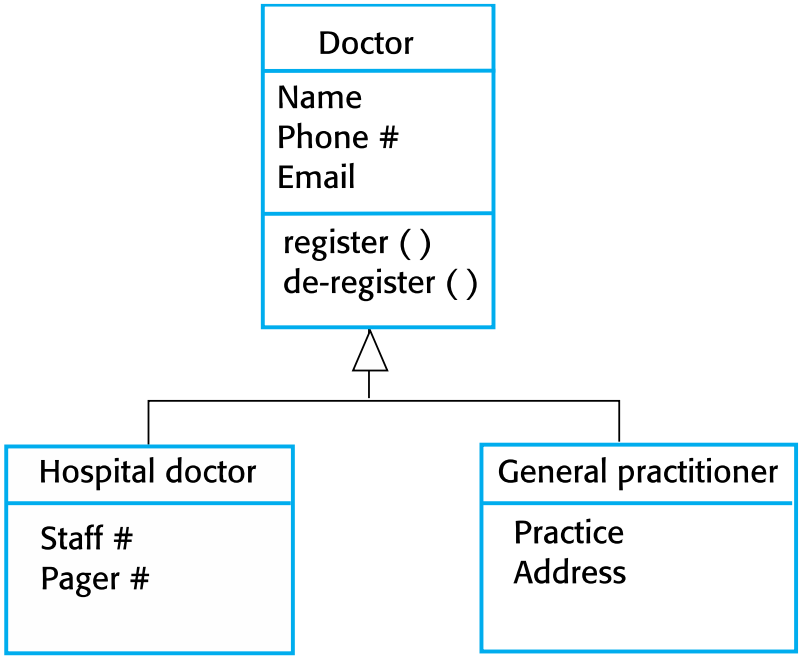
- All doctors, including hospital doctor and general practitioner, have a name, phone number and email.

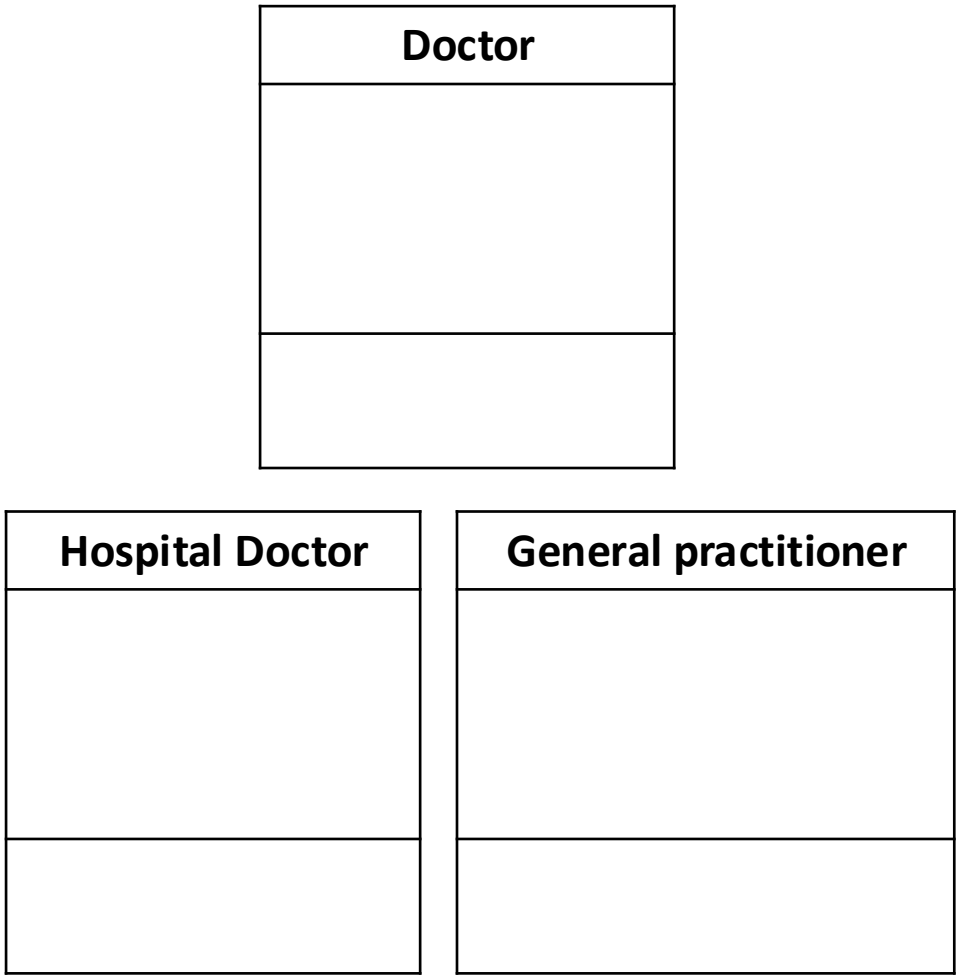
These lower-level classes may include **more specific attributes and operations**.

In this example,

- All hospital doctors have a staff number and carry a pager.
- General practitioners do not have these attributes, as they work independently, but they have an individual practice name and address.

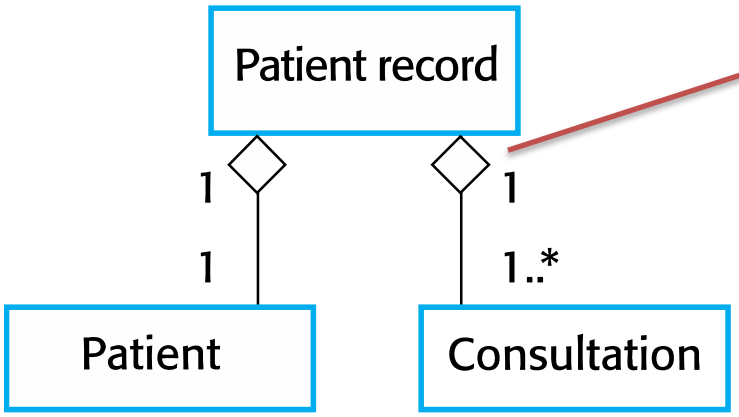
How can we maintain the same usage in the UML diagram **without using generalization**?





Object class aggregation models

- An aggregation model shows how classes that are collections are **composed of** other classes.



Aggregation - one object (the whole) **is composed of** other objects (the parts). It forms a **"part of"/ "has a"** relationship.

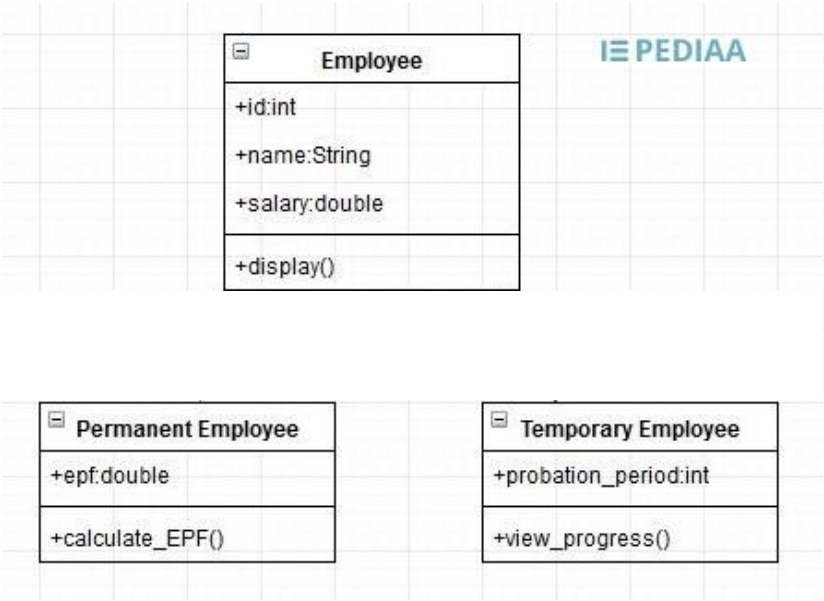
In this example,

- A patient record is composed of a Patient and an indefinite number of Consultations.

Generalization vs. Aggregation: Understanding the Differences

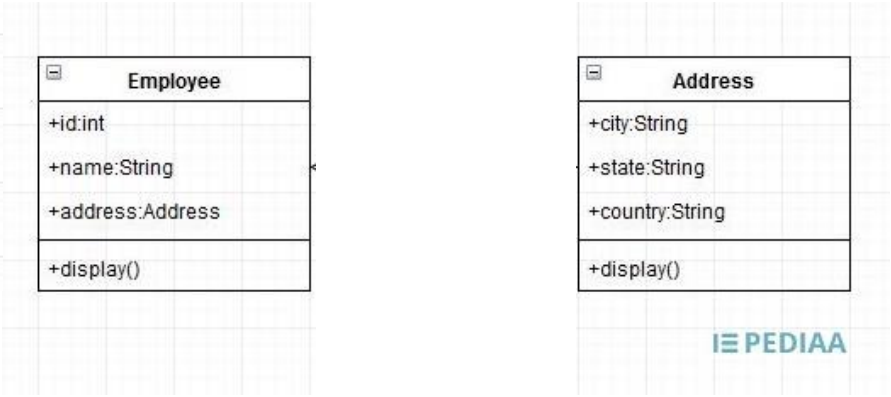
Generalization / Aggregation

Reason: _____

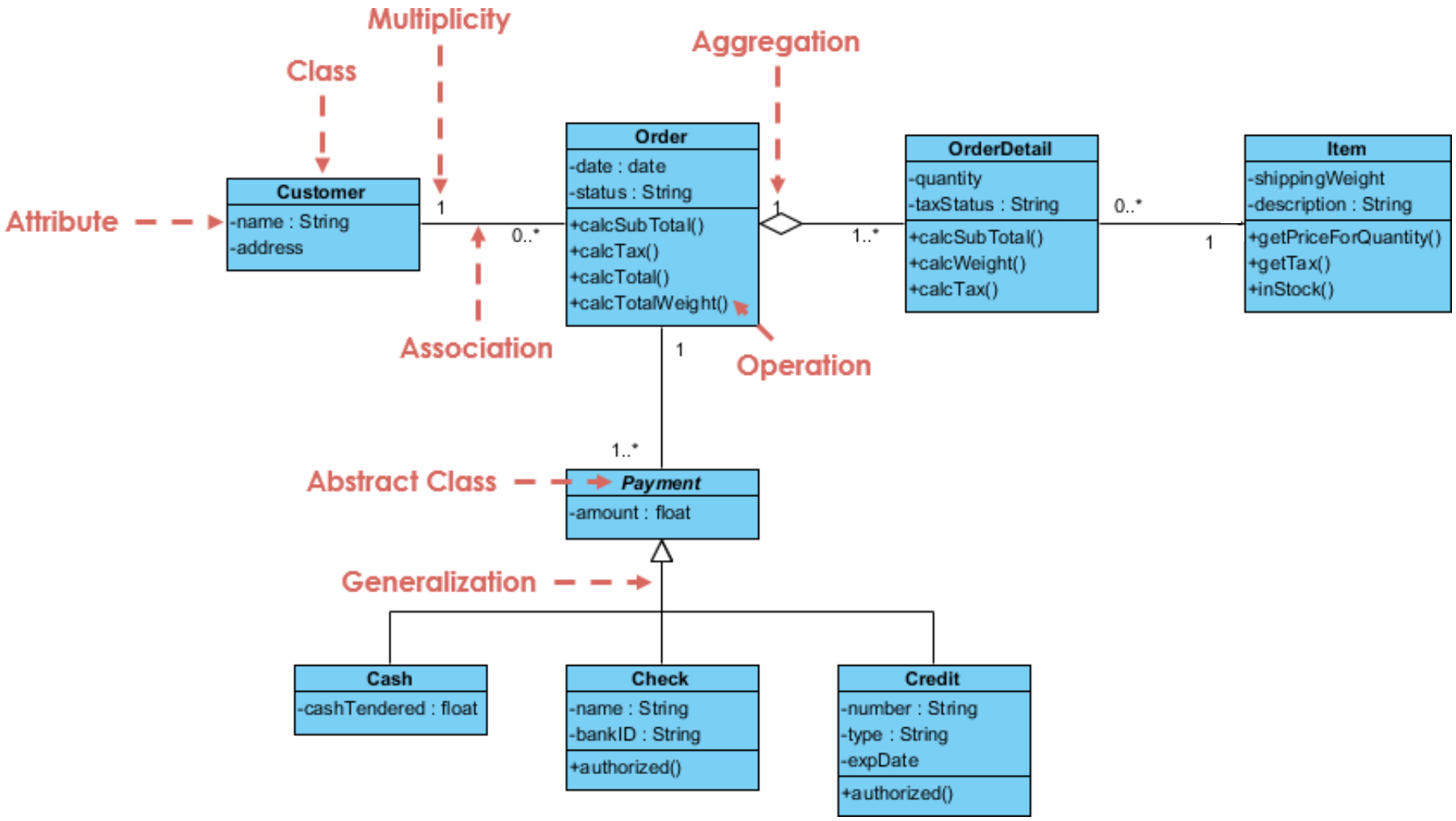


Generalization / Aggregation

Reason: _____



All notations in a Class Diagram: Order System



Text-Based UML: Mermaid

- You **type the diagram as text**; the tool draws the picture.
- Mermaid supports class, sequence and state diagrams. **GitHub shows them directly** in README and other Markdown files.
  - So your project repo can show **up-to-date diagrams** with no plug-in and no image files.
- One gap: Mermaid has **NO use case diagram**. If you ask AI for one, it will still give you a fake one (you will see this in W3).
- Why text? It works with Git: **version control, diff and code review**. AI tools can also write it well.
- Always **review AI-generated diagrams**: do they match the user stories, the notation, and the other diagrams? AI makes the same mistakes you saw today, only faster (hands-on in W3).

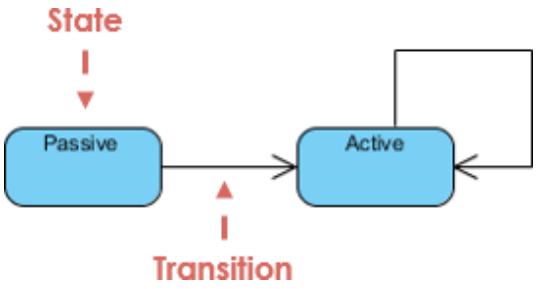
Behavioral models

Behavioral models

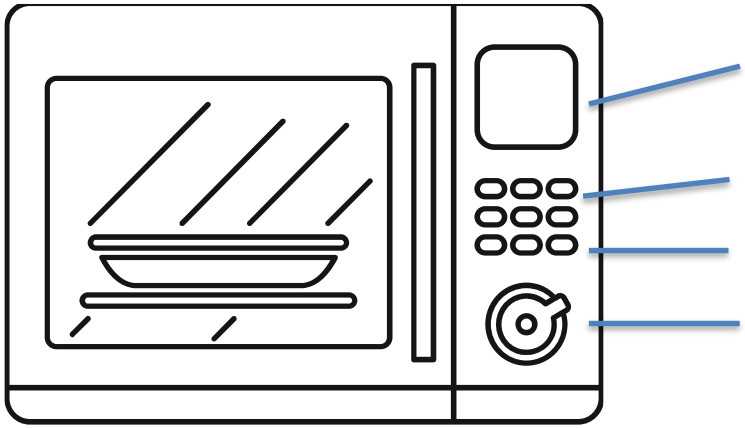
- **Behavioral models** are models of the **dynamic behavior of a system** as it is executing. They show what happens or what is supposed to happen when a system **responds to a stimulus from its environmen**t. There are two types of stimuli:
  - **Data** - Some data arrives that has to be processed by the system.
  - **Events** - Some event happens that triggers system processing. Events may have associated data, although this is not always the case.
- **Event-driven modeling**
  - **Real-time systems** are often event-driven, with minimal data processing. For example, a landline phone switching system responds to events such as 'handset activated' by generating a dial tone.  
  - Event-driven modeling shows how a system responds to external and internal events.

State machine models

- These model the behaviour of the system in response to **external and internal events**.
- They show the **system's responses to stimuli** so are often used for modelling real-time systems.
- **State machine models** show system **states as nodes** and **events (or transition/trigger) as arcs** between these nodes. When an event occurs, the system moves from one state to another.



Partial descriptions of a simple microwave oven



an alphanumeric display to display various alerts and warning messages

a numeric keypad to input the cooking time

a start button and a cancel button

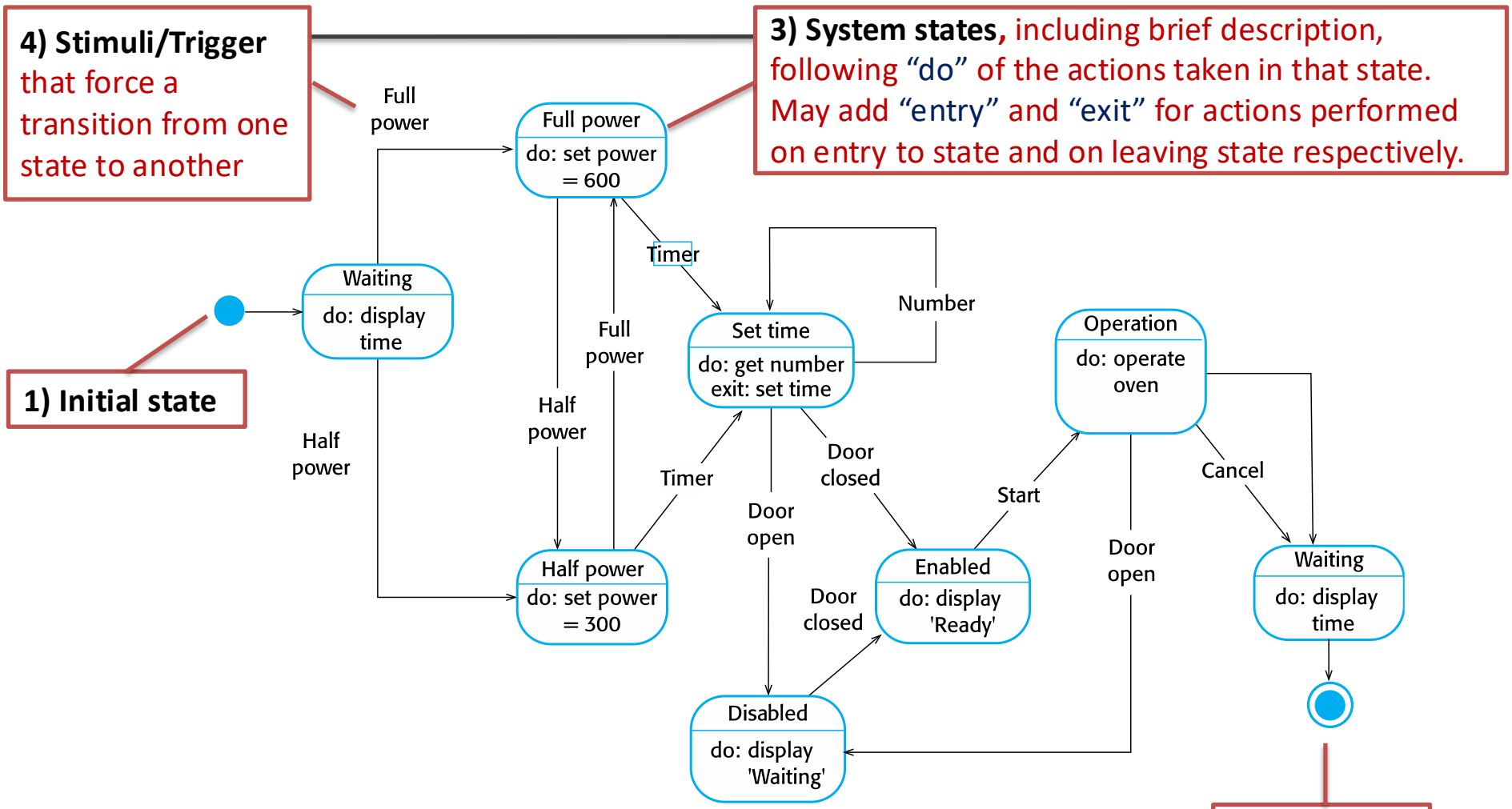
a switch to select full or half power

- Sequence of actions in using the microwave oven
  1. Select the power level (either half power or full power).
  2. Input the cooking time using a numeric keypad.
  3. Press Start and the food is cooked for the given time.
- For safety reasons, the oven should not operate when the door is open. On completion of cooking, a buzzer is sounded.
- …

What do the UML notations mean?

4) Stimuli/Trigger that force a transition from one state to another

3) System states, including brief description, following "do" of the actions taken in that state. May add "entry" and "exit" for actions performed on entry to state and on leaving state respectively.



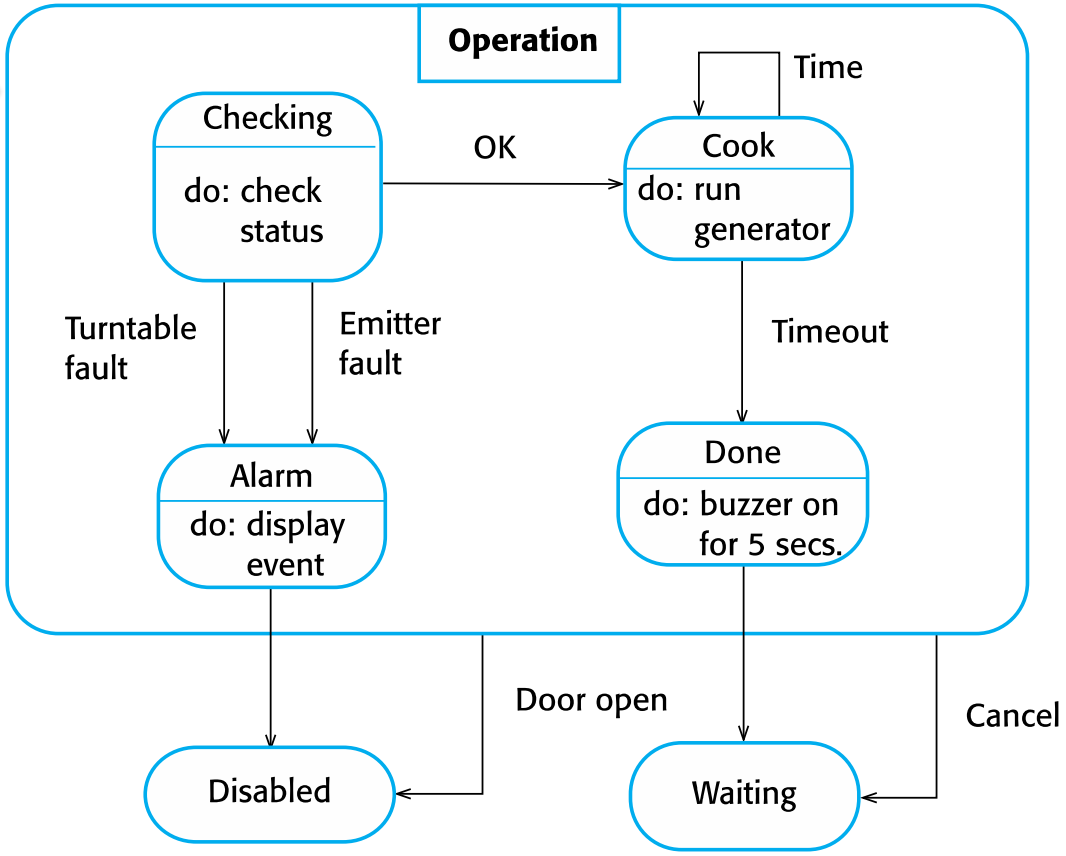
1) Initial state

2) Final State

Example: State Machine diagram of a microwave oven

Microwave oven operation

Superstate (e.g. Operation) containing a set of **substates** (e.g. Checking, Cook, Alarm, Done)

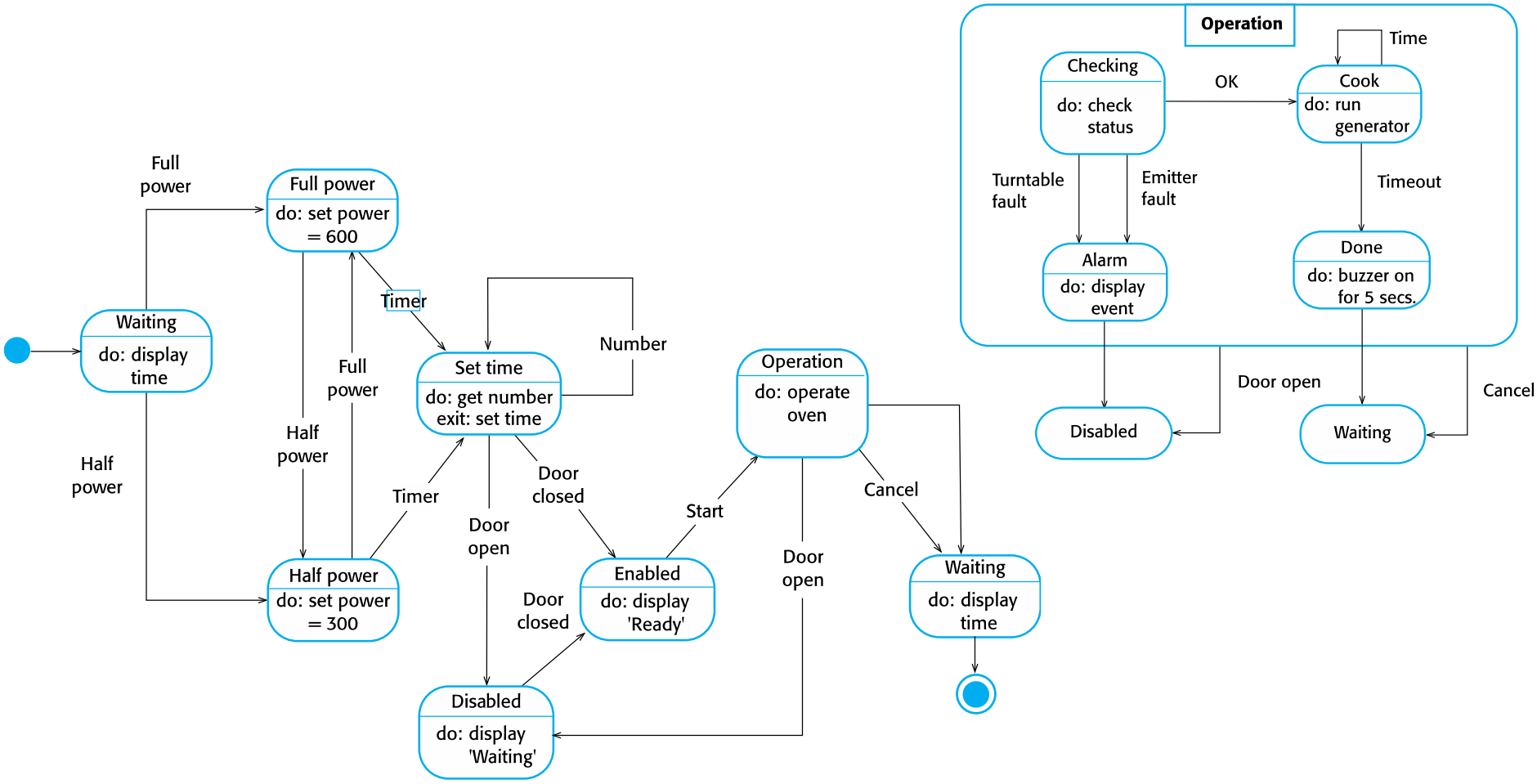


States and stimuli for the microwave oven

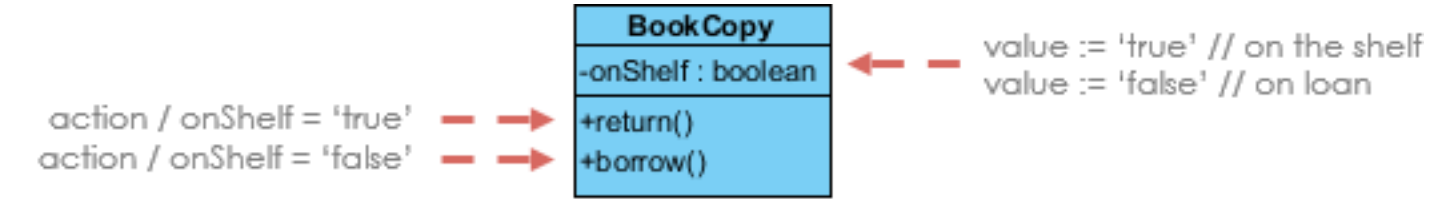
State	Description
Waiting	The oven is waiting for input. The display shows the current time.
Half power	The oven power is set to 300 watts. The display shows 'Half power'.
Full power	The oven power is set to 600 watts. The display shows 'Full power'.
Set time	The cooking time is set to the user's input value. The display shows the cooking time selected and is updated as the time is set.
Disabled	Oven operation is disabled for safety. Interior oven light is on. Display shows 'Not ready'.
Enabled	Oven operation is enabled. Interior oven light is off. Display shows 'Ready to cook'.
Operation	Oven in operation. Interior oven light is on. Display shows the timer countdown. On completion of cooking, the buzzer is sounded for five seconds. Oven light is on. Display shows 'Cooking complete' while buzzer is sounding.

Stimulus	Description
Half power	The user has pressed the half-power button.
Full power	The user has pressed the full-power button.
Timer	The user has pressed one of the timer buttons.
Number	The user has pressed a numeric key.
Door open	The oven door switch is not closed.
Door closed	The oven door switch is closed.
Start	The user has pressed the Start button.
Cancel	The user has pressed the Cancel button.

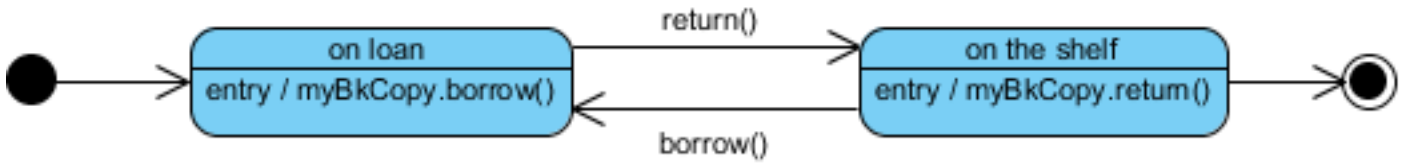
Any problem(s)?



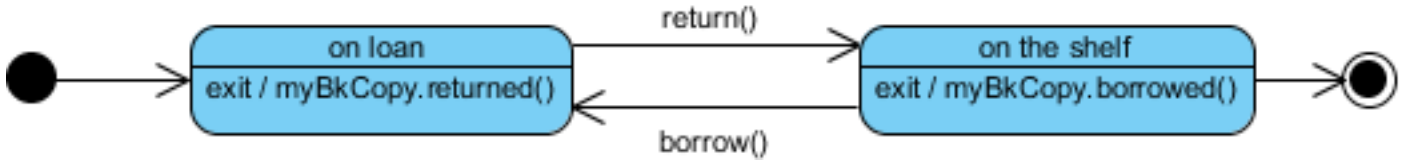
Example of a state machine diagram derived from a Class - "BookCopy"



Option 1:



Option 2:



(Ref: Visual Paradigm)

References

- Textbook: Sommerville, 10th ed., Chapter 5 (System Modeling).
- OMG UML specification
- Mermaid docs | draw.io (visual editing)
- GitHub Docs - creating diagrams in Markdown